



Recent advances in selective separation technologies of rare earth elements: a review

Ziying Chen^{a,b}, Zhan Li^c, Jia Chen^a, Parashuram Kallem^{a,d}, Fawzi Banat^d, Hongdeng Qiu^{a,e,f,*}

^a CAS Key Laboratory of Chemistry of Northwestern Plant Resources and Key Laboratory for Natural Medicine of Gansu Province, Lanzhou Institute of Chemical Physics, Chinese Academy of Sciences, Lanzhou 730000, China

^b College of Chemistry and Chemical Engineering, Xinjiang Normal University, Urumqi 830054, China

^c Frontier Science Center for Rare Isotopes, Lanzhou University, Lanzhou 730000, China

^d Center for Membranes and Advanced Water Technology (CMAT), Khalifa University of Science and Technology, P.O. Box 127788, Abu Dhabi, United Arab Emirates

^e College of Chemistry and Chemical Engineering, Gannan Normal University, Ganzhou 341000, China

^f College of Chemistry, Zhengzhou University, Zhengzhou 450001, China

ARTICLE INFO

Editor: Dr. GL Dotto

Keywords:

Rare earth elements
Separation technology
Selective Separation
Adsorption method
Review

ABSTRACT

With the continuous improvement of technological level, the potential of rare earth elements (REEs) called "industrial gold" in various fields has been continuously explored, and some rare earth elements are regarded as very valuable additives in the high-tech field. With the rapid development of the rare earth industry in recent years, the demand for REEs has continued to rise. Because neighboring rare earth elements have extremely similar physical and chemical properties, in industrial applications, the purity of rare earth elements has high requirements, so the separation between rare earth elements is particularly important. Over time, people have developed a variety of techniques for the separation and recovery of rare earth elements, commonly used including chemical precipitation, ion exchange, solvent extraction, membrane separation, and adsorption, etc. At present, there are few reports on the comprehensive progress of all separation and purification technologies for a single rare earth element. This article mainly summarizes the latest developments in the separation of rare earth elements, evaluates the advantages and disadvantages of various process technologies, and prospects the development prospects of rare earth element separation technologies.

Abbreviations: 3D-GO-PER, Three-dimensional graphene oxide pentaerythritol; [C₄ mim]⁺ [Br]⁻, 1-butyl-3-methylimidazole bromide; [P6,6,6,14] [SOPAA], [triethyl(tetradecyl)phosphonium] [sec-octylphenoxycetic acid]; P(FPMAM-co-MAA), Poly (ferrocene propyl methacrylamide-methacrylic acid); AcPhos, Acetamide phosphonic acid; AC, Activated carbon; BPAOA, 2-[benzyl(propan-2-yl) amino] – 2-oxoacetic acid; BEAOA, 2-[benzyl(ethyl)amino] – 2-oxoacetic acid; Bif-ILs, Bifunctional ionic liquids; BLM, Bulk liquid membranes; CSEREOX, Chemical separation program for rare earth oxalates; CNCs, Cellulose nanocrystals; CNTs, nanotubes; CGC, GO/cellulose composite; CZ, corn zein; COF, Covalent organic framework; CP, Coordination polymer; CMPO, Carbamoyl phosphine oxide; DBOA, 2-(dibenzylamino) – 2-oxoacetic acid; DGA, Diglycolamide; DESs, Deep eutectic solvents; DETA, Diethylene triamine; DTPA, diethylene triamine pentaacetic acid; ELM, Emulsion liquid membranes; ESPLM, Electrostatic pseudo liquid film; EDTA, Ethylenediaminetetraacetic acid; EW, Egg white; ED, Electrodialysis; GIMs, Gd (III)-imprinted membranes; GO, Graphene oxide; GA, Gum Arabic; GICNTs, Gd (III) embossed carbon nanotubes; HDEHP, Di(2-ethylhexyl) phosphoric acid; HFSLM, Hollow fiber supported liquid membranes; HGF, High-gluten flour; Hfod, Fluorinated β-diketones; ICP-MS, Inductively coupled plasma mass spectrometry; INET-3, (2,4,40-trimethylpentyl) phosphonic acid; ILs, Ionic liquids; MOF, metal framework; NDNG, Nanoporous graphene membrane; NdFeB, neodymium-iron-boron; PC-88A, 2-ethylhexylphosphonic acid mono-2-ethylhexyl ester; PIM, Polymer inclusion membrane; PVDF, Poly (vinylidene fluoride); PDA, Phenanthroline diamide; PAM, Polyacrylamide; Prop Phos, Propionamide phosphonic acid; PDPA, Phenylidioxidamine; PAN, 1-(2-Pyridylazo) 2-naphthol; POCl₃, Phosphite phosphorus oxychloride; PAA-S, Poly (acrylic acid) – silica; PAA-S-HNFs, Poly (acrylic acid) – silica hydrogel nanofibers; REEs, Rare earth elements; R₄NCy, Bis (2,4,4-Tri-methylpentyl) ammonium; R₄ND, Trioctyl methyl bis(2-ethylhexyl) ammonium phosphate; RTILs, Room temperature ionic liquids; SIRs, Solvent impregnated resins; SLM, Supported liquid membranes; SAMMS, self-assembled monolayers on mesoporous supports; SF, Separation factor; TBP, Tributyl phosphate; TBP-HNO₃, Tributyl phosphate-nitric acid; TSILs, Task specific ionic liquid; TEMPO, 2,2,6,6-tetramethylpiperidine-1-oxyl; TAPA, Tris(4-aminophenyl) amine; TMS-EDTA, N-[(3-trimethoxysilyl) propyl] ethylenediaminetriacetic acid; UIO-66, C₄₈H₂₈O₃₂Zr₆; VSC, Valence Selective Crystallization; WMS, Wrinkled mesoporous silica; WMC, Wrinkled mesoporous carbon.

* Corresponding author at: CAS Key Laboratory of Chemistry of Northwestern Plant Resources and Key Laboratory for Natural Medicine of Gansu Province, Lanzhou Institute of Chemical Physics, Chinese Academy of Sciences, Lanzhou 730000, China.

E-mail address: hdqiu@licp.cas.cn (H. Qiu).

<https://doi.org/10.1016/j.jece.2021.107104>

Received 27 October 2021; Received in revised form 19 December 2021; Accepted 27 December 2021

Available online 29 December 2021

2213-3437/© 2021 Elsevier Ltd. All rights reserved.

1. Introduction

In Mendeleev's periodic table, 17 elements, including lanthanides, yttrium and scandium, are collectively referred to as rare earth elements [1,2]. Due to their different physical and chemical properties and subtle differences in ionic radius, they can be broadly divided into light rare earths and heavy rare earths. The elements from lanthanum (La) to europium (Eu) are considered light rare earths, while the others from gadolinium (Gd) to lutetium (Lu) are considered heavy rare earths [3]. As a strategic resource, rare earths have become indispensable raw materials in the fields of national defense, energy, glass ceramics and agriculture [4].

With the continuous development of science and technology worldwide, the potential of rare earth elements in various fields has been continuously explored. Adding rare earths to mechanical castings can help improve the low-temperature toughness and mechanical properties of materials; as luminescent materials, rare earths not only have good light efficiency and a long life but are also more energy saving than incandescent lamps. It is worth noting that super-permanent magnet materials made of neodymium, praseodymium, dysprosium, etc. are 4–10 times more magnetic than ordinary permanent magnet materials, they are widely used in high-tech fields such as TV, medical equipment, maglev trains and the military industry [3–5]. By using a small number of rare earths in metallurgy, petroleum industry, chemical industry, and other fields, the performance of various products can be greatly improved. This increase in performance can effectively improve production efficiency [6]. Therefore, the demand for rare earths in various fields continues to increase.

People's demand for rare earths continues to increase, but the natural rare earth reserves are limited, coupled with imperfect rare earth industrial post-processing techniques in individual areas, which has resulted in the emission of a large amount of secondary raw materials containing rare earths. It is not only a huge waste of rare earths but also a huge burden on the environment [7]. Moreover, some secondary raw materials contain trace amounts of rare earths (e.g., Gd), but even after they accumulate in organisms, they are still potentially harmful to human health [8]. Researchers have turned their attention to the recovery of rare earths from secondary raw materials such as rare earth tailings, industrial waste, magnetic waste, and electronic waste, which has environmental and resource reuse benefits [9–11].

In natural mineral resources and rare earth secondary resources,

many other heavy metal elements are usually associated, but high-purity rare earths are required in industrial applications, so the separation of rare earth elements has received attention in recent years [12,13]. In the process of extracting or recovering rare earths from rare earth minerals and secondary resources, rare earths usually exist in the form of insoluble oxalates and chlorides. To obtain a single rare earth element, it must first be separated from the precipitate of non-rare earth elements, and then the single rare earth element can be separated [14]. The second is the separation of rare earth elements. However, the physical and chemical properties of adjacent rare earth elements are similar, and it is not easy to separate and purify a single rare earth element [15].

Rare earths usually exist in single or binary mixed forms [16]. In order to obtain a single rare earth element, researchers have conducted a lot of research [17]. Pyrometallurgical and hydrometallurgical processes are commonly used in the reports [11]. Although pyrometallurgy reduces the use of organic solvents, the melting of metals requires high-temperature conditions, which not only consume a large amount of energy but also require the addition of chemicals for high-temperature treatment, which inevitably involves the generation of waste and complex processes. Therefore, we believe that it is not suitable for purification of individual rare earth elements [18]. Currently, hydrometallurgy is a mature technology for rare earth separation. In order to separate a single rare earth element from minerals or calcination residues, the following three stages must be passed (as shown in Fig. 1). In the first stage, the mixed solution containing rare earths is first chemically leached with an aqueous inorganic acid (e.g., HCl, H₂SO₄, or HNO₃); in the second stage, the leaching solution is suitable for separation and purification technology to extract single or mixed rare earths, depending on the different production requirements; in the third stage, the rare earths are recovered by precipitation [19]. However, the hydrometallurgical process requires the use of a large amount of organic solvent, so further research is needed to solve the problem of environmental compatibility.

Currently, it is still an urgent task to develop rare earth separation technologies and materials that are highly efficient, environmentally friendly, and suitable for industrial production. The separation of a single rare earth element is based on the similar but slightly different ion radii of the rare earth elements. Over time, people have developed a variety of techniques for separating and recovering rare earth elements. Commonly used methods include chemical precipitation, ion exchange, solvent extraction, membrane separation, and adsorption, etc. [20–23].

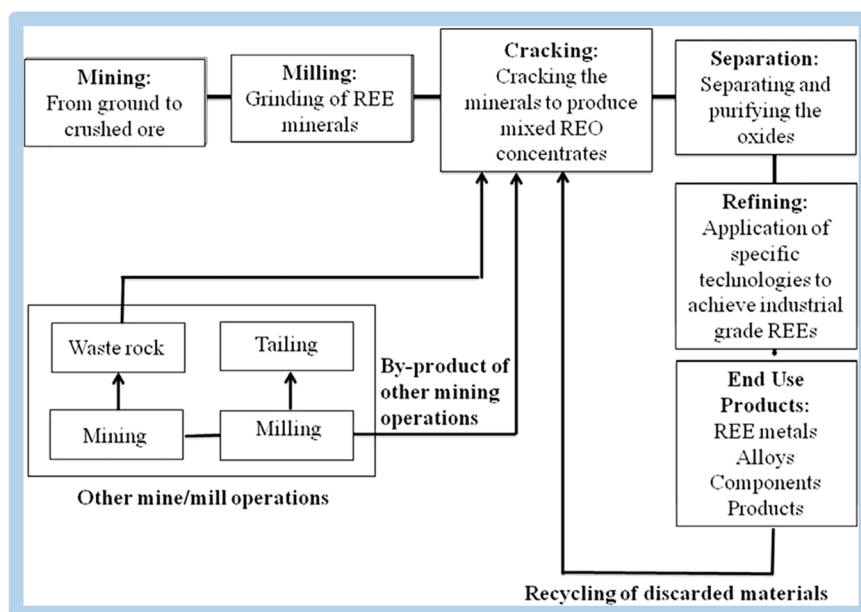


Fig. 1. Steps of rare earth element (REE) ore concentration procedure and recycling [10].

Recently, Mehdi et al. described in depth the separation of rare earth elements by a variety of adsorbent materials, and provided valuable insights into the future development direction of adsorbent materials [11]. Liu et al. provided an in-depth description of the latest progress in the extraction and separation of high purity heavy rare earth elements (HREEs) [18]. Xie et al. summarized a variety of solvent extractants used in the separation of rare earth elements and the typical structure of the extraction process [20]. Li et al. reviewed the latest developments in membrane technology applied to rare earth separation, put forward the limitations of each membrane separation technology at present, and made prospects for its future research directions [21]. Opare et al., through the combination of agricultural mining methods and separation technology, provided a comprehensive overview of the technological challenges and potential business opportunities of the existing rare earth separation technology [24]. In this article, we provide a comprehensive overview of the technologies that have the potential to selectively separate individual rare earth elements, give a brief evaluation of each method, and look forward to future research directions. We hope that this review can provide a useful reference for the separation and purification of rare earths in actual production.

2. Separation techniques

2.1. Step by step method

Since the discovery of a single rare earth element, the separation and purification of rare earth elements has been carried out gradually over the last hundred years. In this process, the difference in solubility of rare earth elements is exploited, and various chemical reactants are added to perform separation by crystallization and precipitation [25]. Based on precipitation methods, rare earth ions can form complexes with different ligands, which can help us to quickly separate rare earth mixtures [26]. Prodiu et al. applied a chemical separation program for rare earth oxalates (CSEREOX) to realize the recovery of Nd-Fe-B magnets and the extraction of Dy [27]. This procedure is low-cost and environmentally

friendly. In 2018, Cole et al. performed some related work on the properties and structure of ligands bound to rare earth elements. The aryl backbone of $H_3TriNOx$ is replaced with methoxy groups, which affects the REE ($TriNOx$) separation system [18]. This work provides a basis for the influence of different ligand environments on the separation of rare earths. In 2020, it was also demonstrated that the electronic effect of aryl substituents affects the association and dimerization with rare earths (neodymium/dysprosium), which increases the difference in solubility of rare earth elements in solvent extraction, achieving the highly selective separation of a single rare earth element [28]. Although the separation ability of rare earths is improved, due to the small solubility difference between adjacent rare earths, the separation process of the stepwise method has a long operation cycle, large losses of rare earths, and poor separation effects, and it has not been widely used [29].

It has been reported that the radius difference between adjacent rare earth ions is only approximately 0.03 Å, so it is extremely challenging to separate rare earths based on the size difference. The study used inexpensive camphor acid and lanthanides to synthesize organic rare earth metal framework materials with various structures and properties, revealing for the first time the enormous potential of organic rare earth metal framework materials in separating individual rare earth elements (as shown in Fig. 2a-f) [30]. During the formation of the crystal framework of MOFs, the size of rare earth ions and their subtle changes will enhance the chances of the repulsive force between the connected ligands, showing a high degree of size selective crystallization and can effectively separate a single rare earth element. Based on the method of selective crystallization of borate, Wang et al. improved the difference between trivalent lanthanides and tetravalent thorium to purify radioactive pollutants in rare earths [32]. This work provides ideas for separating metals based on their valence states. Yang et al. proposed a Valence Selective Crystallization (VSC) strategy using Ce as a target ion to selectively crystallize from the mixed rare earth solution into a preset Ce(IV)-MOF in a green and gentle manner (Fig. 2g). The accurate separation of Ce in the mixed solution, the purification efficiency is as high as 99.97%, and the scale-up experiment still has good selective

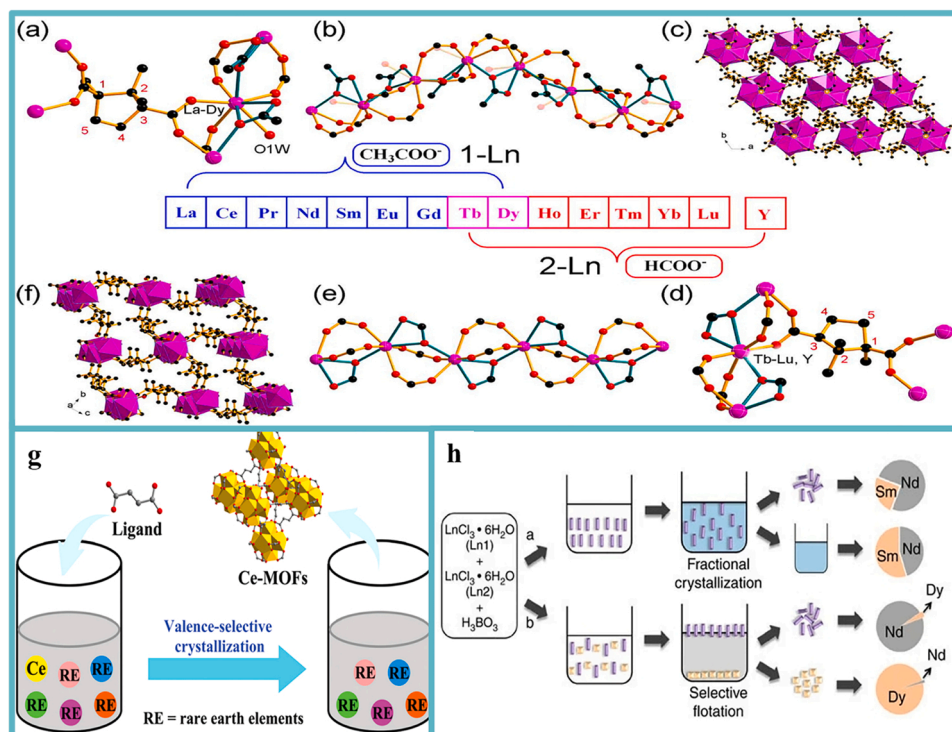


Fig. 2. (a-f) Schematic diagram of selective crystallization of lanthanides [30]. (g) Schematic diagram of precise separation of Ce^{4+} in rare earth solution using valence selective crystallization [12]. (h) Schematic diagram describing the selective crystallization and separation of lanthanides [31].

separation of target ions, which are expected to be applied to industrialization [12]. This work is not only environmentally friendly and energy-saving, but also able to accurately separate mixed Ce, but most of the rare earth element valence states exist in trivalent form, and only Ce can show tetravalent valence, which is more suitable for accurate capture of tetravalent ions. Yin et al. achieved the quantitative separation of a single rare earth element by controlling the reaction kinetics of the binary rare earth models Nd/Sm and Nd/Dy and proved that the crystallization of borate allows specific identification of rare earth elements (Fig. 2f) [31]. This work breaks the multiple repetitive operations necessary in the previous stepwise separation of a single rare earth element. In this work, different ions react with the same ligand to grow different crystals. These crystal mixtures can be further separated by flotation. This is another important development in the separation of rare earth elements based on the size crystallization method.

Since the size difference of rare earth elements is very small, the traditional step method requires dozens or even hundreds of dissolution-precipitation operations to obtain a single rare earth element. Considering the production cycle and cost, it is not suitable for industrial production. Compared with the traditional stepwise method, the size-selective crystallization method based on MOFs is simple in process, easy to operate, and can obtain a single rare earth element with high purity. This technology only requires one step of operation, and the quantitative separation of a single rare earth element is achieved by controlling the reaction kinetics. Finally, the crystalline products of different rare earth elements are separated by flotation [30–32]. This technology provides a new idea for the separation of rare earths based on ion size. However, the size-selective crystallization mechanism of MOFs needs further study.

2.2. Ion exchange methods

The step-by-step method has poor separation effect and low purity, but the ion exchange method can make up for this defect, so it came into being. In the 1960s, before the advent of solvent extraction, ion exchange technology was the only viable method to separate rare earths on a large scale [33]. The ion exchange resin is used as the exchanger (strongly acidic cation exchanger and strong basic anion exchange resin) to separate a single rare earth element. When the ion exchanger comes into contact with the solution, an exchange reaction occurs between the two. The ion in the solution with the same charge sign as the exchangeable ion of the resin diffuses into the inside of the exchanger, and the ion of the exchanger is transferred into the solution. At this time, the resin skeleton and fixed ionic groups do not change during the exchange [34,35]. The use of different ion exchange resins offers unlimited possibilities for the functionalization of extraction resins. For ion exchange methods, the key to improving the selective separation performance of rare earths lies in the selection of ion exchangers and the conditions of adsorption and leaching. The nature of the exchange group, the degree of crosslinking, and the size of the exchange capacity have an important influence on the exchange process [36].

In 1978, the physical impregnation technology was reported. This technology can evenly distribute the metal extractant on the surface of the polymer carrier to obtain the metal ion extraction resin. Compared with conventional ion exchange resins, the metal ion complex and release rate are better. This technology can not only reduce the use of organic solvents, but also has high selectivity and high ion exchange performance [41]. However, solvent-impregnated resins (SIRs) are easily lost in the carrier matrix in industrial applications. To solve this problem, Yamada et al. impregnated the support material with the

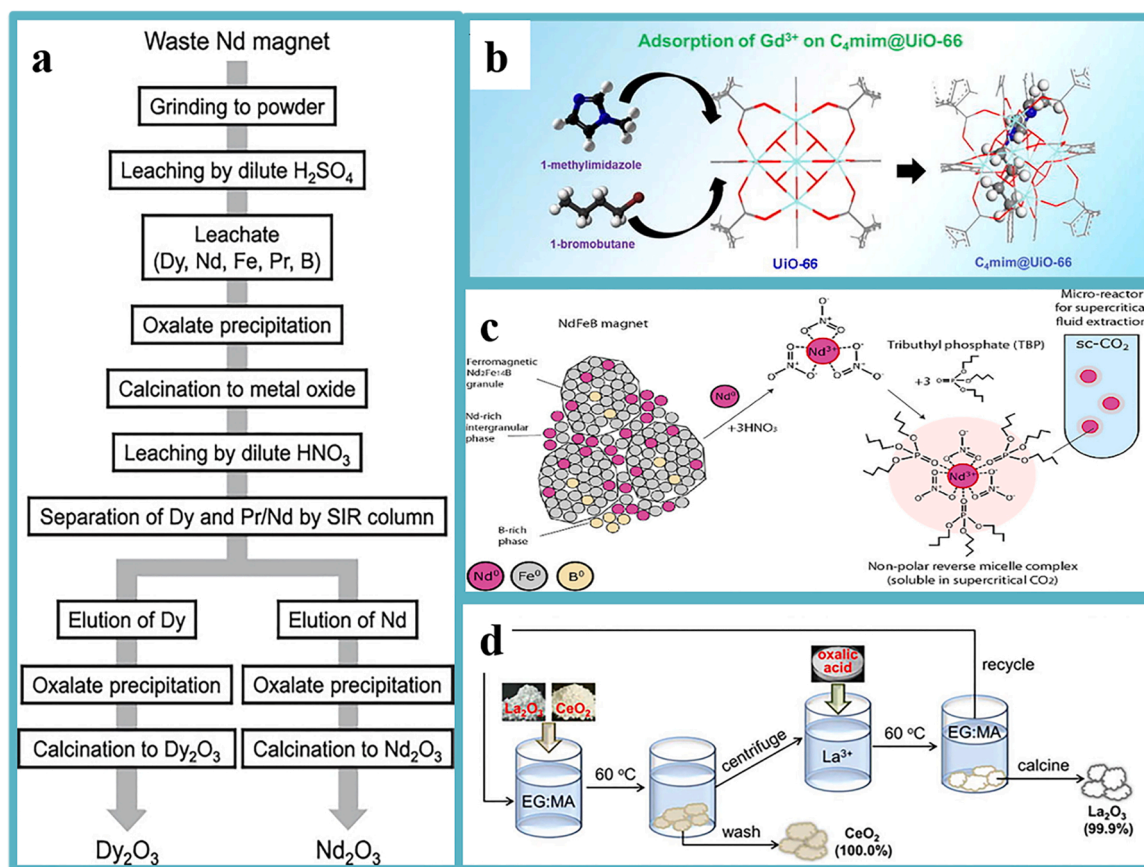


Fig. 3. (a) Complete flowchart of the recovery process of Dy and Nd from the waste Nd magnet [37]. (b) Capture of Gd^{3+} in solution by UiO-66 coated with ionic liquid [38]. (c) Illustration of the supercritical carbon dioxide (sc-CO_2) extraction model [39]. (d) Schematic diagram of the separation process of rare earth oxides [40].

extractant 2-ethylhexylphosphonic acid mono-2-ethylhexyl ester (PC-88A) and then coated it with glutaraldehyde cross-linked polyvinyl alcohol to prevent the loss of the extractant. As shown in Fig. 3a, it is used as a chromatographic filler to recover Dy and Nd in waste neodymium magnets. The purities of Dy and Nd can reach 98.5% and 91.6%, respectively [37]. However, the ion exchange resin itself is an organic substance, and usually after a period of use, the mechanical properties become worse, and when the functional group reaches saturation, the separation performance will be greatly reduced. In addition, the reusability of the resin is not good, so the development is limited.

Ionic liquids have aroused great interest of researchers because of their green environmental protection and functional designability. Ionic liquids are solvents composed of only ions, and their physical and chemical properties can be designed by changing anions and cations. In recent years, some researchers have focused on the separation of rare earths by new ionic liquid extraction resins (common resins, DGA, DN, etc.) [42]. Different rare earth ions and ionic liquid anions/cations, or both are chelated with anions and cations. The chelated rare earth element enters the ionic liquid from the aqueous solution, and finally the ionic liquid containing the rare earth ions continuously stripped to separate the REE. However, under certain conditions, both the cation and anion components of the ionic liquid can enter the water phase, resulting in the loss of the ionic liquid, which is not conducive to the recycling of the ionic liquid. The combination with polymer materials effectively reduces the loss of ionic liquid. The ionic liquid extractant (P507, [A336], etc.) is uniformly loaded on inorganic/organic polymer materials by cross-linking reaction to form an ionic liquid film, which combines ion exchange resin with chromatography and ICP-MS, etc., to separate solutions with low rare earth content [35,42,43]. Compared with traditional resin materials, ionic liquid extraction resin not only increases the carrier contact area, but also has low cost and high recovery rate. Moreover, due to the combination with polymer materials, the loss of ionic liquid in the water phase is greatly reduced. Ahmed et al. used the ship-in-a-bottle method to embed the ionic liquid 1-butyl-3-methylimidazole bromide ($[C_4\text{mim}]^+[\text{Br}]^-$) in a hollow cage with a stable structure UiO-66 [38]. Since the presence of ionic liquid provides more adsorption sites for the adsorbent, this adsorbent not only shows high adsorption capacity for Gd^{3+} but also has a selectivity of up to 82% for metal ions (as shown in Fig. 3b). This provides us with new ideas. In order to reduce the cost of resin replacement, perhaps the combination of ionic liquids and polymer materials that are easy to recycle is the future development direction of ion exchange technology.

The ion exchange method can separate a variety of elements in one operation, has the advantages of low pollution and high product purity, but has a long production cycle. Considering that the resin material is poor in regeneration and usability, continuous processing capacity is limited, resulting in high process costs. It is not currently the mainstream technology of rare earth separation. Although ion exchange technology has been almost replaced by solvent extraction in the actual separation of rare earth elements, the high purity rare earth elements obtained by ion exchange technology make this technology necessary for further research [41]. We believe that only by researching ion exchange technology that is highly selective and reusable for rare earths can it be applied to industrial production. Perhaps combining the ion exchange method with other separation technologies can better utilize the advantages of this existing ion exchange method, such as ionic liquids replacing the environmentally-friendly extractants in the solvent extraction method, and polymer materials coated with ionic liquids [37, 44–46].

2.3. Solvent extraction

Compared with the ion exchange method, the solvent extraction method can be used for the rapid and continuous production of rare earth elements. The solvent extraction method is suitable for the extraction and separation of rare earths and takes advantage of the

different solubility of the various rare earth complexes in the water and organic phases [47]. In the separation processes of multi-component rare earths, the mixture can generally be pre-separated by the tetrad effect between the lanthanides and then further purified [29]. Due to its large processing capacity, good selectivity, high product purity and continuous operation, solvent extraction has become a mature industrial technology. Three types of extractants are known for the separation of rare earths: cation exchangers (acidic extractants) [48], solvated extractants (neutral extractants) and anion exchangers (alkaline extractants) [49,50]. At present, organic phosphonic acid and organic carboxylic acid extractants are commonly used for the separation of rare earths. As shown in Table 1, we summarized the separation and purification of rare earth elements by some extraction solvents recently.

Bie et al. investigated the separation of thorium and rare earths using three new oxyacetic acid extractants, 2-[benzyl(propan-2-yl) amino]–2-oxoacetic acid (BPAOA), 2-[benzyl(ethyl)amino]–2-oxoacetic acid (BEAOA), and 2-(dibenzylamino)–2-oxoacetic acid (DBOA) [51]. This work solved the problem of secondary entrainment in the hydrochloric acid system. Peng et al. used relatively environmentally friendly bis-glycolamide DGAs with different alkyl chains as extractants to better extract and separate heavy rare earth ions [52]. The extractant used in this research saves a lot of organic reagents and has good selective separation ability for heavy rare earth ions. Liu et al. reported an asymmetric dialkylphosphonic acid extractant (2,4,40-trimethylpentyl) phosphonic acid (INET-3), which has a stronger extraction capacity than Cyanex 923. The recovery purity of this extractant for neodymium can reach 99% in only three countercurrent extraction steps, and the recovery purity for Dy can reach 98% in four countercurrent extraction steps [53]. Although this extractant has a good extraction effect, it is not environmentally friendly.

Tributyl phosphate (TBP), the most common neutral extractant, is highly soluble in water but easily leads to the loss of organic solvents. Cyanex is a good neutral extractant because of its high efficiency and low water solubility, which has been used for separation and recovery of rare earth elements Ce(III), Dy(III), and Nd(III) [49,54]. To solve the problem that the pH value of the cationic extractant used in industry is not stable enough, Pavón et al. chose an ionic liquid in which no proton exchange occurs during extraction and synthesized Primene 81 R-Cyanex 572 as a countercurrent process for the extraction of rare

Table 1
Separation of rare earth elements by different extraction solvents.

Target REEs	Extraction solvents	Optimum experimental conditions	References
Ce	Cyanex 923	Ce:H ₂ C ₂ O ₄ at 1:≥ 10, T = 25 °C, pH = 1.5 (± 0.2), t = 300 s	[49]
La, Ce, Pr, Nd, Sm, Eu, Gd, Tb, Dy, Ho, Er, Y, Tm, Yb	Cyanex272/peg600/(NH ₄) ₂ SO ₄ -H ₂ O	C (Cyanex272) = 0.5 mol/L, C (PEG) = 20 (w/v)%. M (REE: DTPA) = 3:2. pH = 2.0.	[29]
Nd, Ho	TBP	C (Tributyl phosphate) = (8–80 mM), C(H ₂ O) = (0 ~ 200 mM). T = 60 °C, P = (17 ~ 28 MPa). M(tributyl phosphate: lanthanides) = (1–2).	[47]
Pr, Nd, Eu, Er	Hfod compounds	T = 40 °C, P = 10–25 MPa.	[60]
Th, U	HDEHP	C (HCl) = 4 M, C (HF) = 0.05 M	[61]
Th, Ln	BPAOA, BEAOA, DBOA	C (DBOA) = 0.03 mol/L, C (Th) 0.002 mol/L, pH = 2.97, C (Stripping Acid) = 6 mol/L.	[51]
Y, Ho, Er, Tm, Yb, Lu	[P6,6,6,14][SOPAA]	C ([P6,6,6,14][SOPAA]) = 0.4 mol/L, C (REE) = 0.2 mol/L, pH = 4.5, V (S: F:W) = 6:2:3 Extraction stages = 13, Scrubbing stages = 6.	[56]

earth elements in industrial production. The rare earths in the leachate can be almost entirely stripped and recovered [55]. The saponification reaction of the acidic extractant can effectively improve the separation efficiency of the extractant, but the wastewater generated by the traditional saponification process needs to be treated twice, which increases the process cost for the separation of the rare earths, and the Na^+ and Ca^{2+} plasma generated during the saponification process is difficult to remove.

Dong et al. proposed for the first time a more environmentally friendly saponification process using ionic liquids to avoid the generation of saponification wastewater, and the ion association extraction mechanism is used to prepare [triethyl(tetradecyl)phosphonium] [sec-octylphenoxycetic acid] ([P6,6,6,14] [SOPAA]), which can be used to effectively separate Y(III) from other heavy rare earth elements [56, 57]. In a report by Riaño and his colleagues, the ionic liquid triethyl (tetradecyl) was used as the organic phase for the successful separation and recovery of Nd_2O_3 and Dy_2O_3 with purity of 99.6% and 99.8%, respectively, from waste NdFeB permanent magnets [58]. Padhan and his colleagues, bifunctional ionic liquids (Bif-ILs) with higher extraction capacity for neodymium and praseodymium by increasing the extraction capacity of bifunctional ionic liquids, trioctylmethyl bis (2,4,4-Trimethylpentyl) ammonium (R_4NCy) and trioctyl methyl bis(2-ethylhexyl) ammonium phosphate (R_4ND) were compared to realize the separation and recovery of neodymium and praseodymium in a magnetic separation solution like neodymium-iron-boron [59]. Combination with ion exchange technology avoids the production of a large number of organic pollutants extractants, is more environmentally friendly, and uses ionic liquids to increase the high separation selectivity of rare earth elements.

Supercritical extraction technology has been used to separate rare earth elements [60]. Currently, extraction agents such as lanthanoid(III) complexes with fluorinated β -diketonates (Hfod) compound, tributyl phosphate (TBP), and di(2-ethylhexyl)phosphoric acid (HDEHP) have been investigated for the supercritical extraction of rare earths [60,61]. Zhang et al. used tributyl phosphate-nitric acid (TBP-HNO_3) as a complexing agent and methanol as a co-solvent to improve the interaction between the solvent and the solute (Fig. 3c). The results confirmed the possibility of separating and recovering rare earths from neodymium-iron-boron magnets with supercritical extractants [39]. Compared with conventional solvent extraction methods, supercritical carbon dioxide has low viscosity and high diffusion efficiency, so it can accelerate the reaction rate and reduce the use of organic solvents in the stripping process. However, the selective separation performance of a single rare earth element needs to be further studied.

In recent years, deep eutectic solvents (DESs) have been considered green alternatives to conventional solvents due to their low cost and toxicity. Deep eutectic solvents are a mixture of hydrogen bond acceptors (such as quaternary ammonium salts) and hydrogen bond donors (such as carboxylic acids) [62]. Based on DESs with green characteristics, Chen et al. designed and synthesized a series of novel Brønsted acidic, DESs based on carboxylic acids and polyols by taking advantage of the significant difference in solubility of metal oxides in DESs, and they were used for the separation and purification of binary rare earth oxides, with purification rates of 99.9% and above (Fig. 3d) [40]. This work first introduced the use of DES as a solvent for the dissolution and processing of rare earth oxide compounds. Metal oxides are easy to dissolve in strong acid systems. In this paper, a strong acidic DES is synthesized using the designability of DES. In view of the different dissolution capabilities of rare earth metals at different pH values, the selective separation of rare earth ions is used by controlling the pH of the synthesized DES.

Although solvent extraction is the most suitable commercial technology for rare earth separation, the separation efficiency of this process is limited, and it is necessary to repeat several extractions to achieve high purity [54,57,63]. A large number of extractants are potentially flammable and toxic and cause secondary pollution. Solvent extraction

research is currently focused on optimizing selective extractants and organic solvents to improve separation efficiency and enrichment factors. We believe that the current focus of solvent extraction research should focus on optimizing green extractants with high separation selectivity to improve separation efficiency and enrichment factors. Research on the recovery and reuse process of a large amount of organic solvent waste liquid caused by the current process.

2.4. Membrane separation

Membrane separation is a method in which membrane materials are used as the separation medium. When there is a certain driving force (such as pressure difference, concentration difference or potential difference) on both sides of the membrane, the components selectively pass through the membrane to achieve separation and purification. This method is efficient, simple, energy-saving, and can reduce the loss of organic solvents. It is a relatively green technology for rare earth separation [64]. However, due to the lack of suitable membrane materials, the application of membrane separation technology for rare earth ion purification is limited. In the last decade, the as-reported membrane separation materials known so far can be divided into liquid membranes and non-liquid (solid) membranes. The liquid membranes that have been reported include supported liquid membranes (SLMs) [65,66], bulk liquid membranes (BLMs), hollow fiber-supported liquid membranes (HFSLMs) [67,68], emulsion liquid membranes (ELMs) [69], electrostatic pseudo liquid film (ESPLMs), etc. However, due to the complicated equipment of the liquid membrane separation method and the low stability of the membrane, it is difficult to use it in industrial production [22]. Currently, the separation of rare earth ions by membrane separation methods has been studied on relatively stable non-liquid membranes, mainly focusing on the preparation of polymer inclusion membranes and ion-imprinted polymer membranes [64,70].

The prepared membrane with high selectivity and high permeability can maintain high stability and environmentally friendly and sustainable performance. The polymer inclusion membrane (PIM) has good adsorption and separation performance and is stable. It is one of the most widely used materials for the separation of metal ions. To achieve high-efficiency separation between adjacent rare earth elements, the addition of extractant plays an important role in increasing selectivity [71]. Wang et al. compared the application of task specific ionic liquids (TSILs)-embedded PIMs in the adsorption and separation of metal ions and chose a task-specific ionic liquid Cyphos-IL-104 (Tetradecyl trihexyl phosphine bis-(2,4,4-trimethylpentyl) phosphonate) embedded polymer membrane (PIM) to achieve effective adsorption and separation of adjacent heavy rare earth elements Lu(III) and Yb(III). Compared with the previous PIMs, the adsorption and separation performance are significantly improved, and the membrane surface exhibits better reusability and fouling resistance after the addition of TSIL-Cyphos-IL-104 [72]. Compared with traditional extractants, TSILs have the advantages of low volatility, good selectivity, and high thermal stability, as well as highly efficient separation performance of metal ions. Chen et al. proposed a simple, inexpensive and environmentally friendly polymer coating film poly(vinylidene fluoride) (PVDF) -[A336] [P507] without the addition of plasticizers to enrich the rare-earth element Lu in the filtrate of ion adsorbing mineral deposits. The asymmetric structure of this membrane favors ion migration, and its stability and reusability have also been demonstrated [70]. Unfortunately, the selective separation ability of this membrane material for rare earth elements has not been mentioned.

Graphene oxide-based (GO) separation membranes have always been a hot spot for separation materials due to their high rejection rate, high stability, and controllable interlayer spacing [73]. Tan et al. proposed a new method for doping heteroatoms in nanomaterials. They synthesized a nanoporous graphene membrane (NDNG) with high stability under acidic conditions, in which the pore size and nitrogen doping amount were controlled by multiple confinement strategy

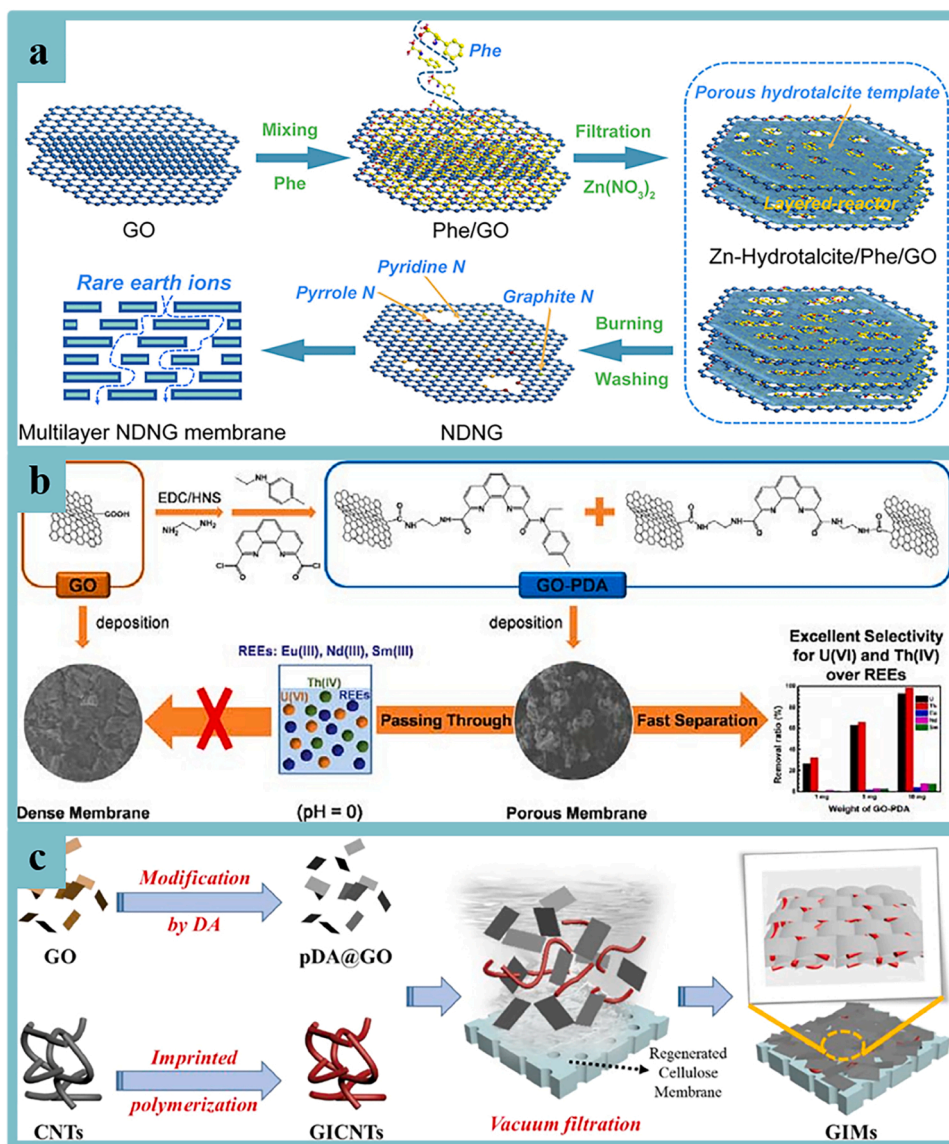


Fig. 4. (a) Schematic diagram of the separation of rare earth elements by an NDNG membrane synthesized by multiple confinement strategy [74]. (b) Schematic diagram of the separation of U(VI), Th(IV) and REEs by graphene oxide modified by PDA [75]. (c) Gd(III)-imprinted membrane (GIM) synthetic route diagram [76].

(Fig. 4a). This membrane material shows high selectivity for the separation of a single rare earth element [74]. However, the reusability of the film material still needs further research. The base film has low binding force and is easily degraded.

The GO base film has a low binding force and is easily degradable. When combined with other substrates on its surface, it can effectively increase the viscosity of the two-dimensional GO base film. Li et al. used phenanthroline diamide (PDA), which is selective to actinides, to functionalize graphene oxide. This method can combine the high adsorption capacity of graphene oxide and the high selectivity of phenanthroline diamide, and then further prepare membranes from this material. The membrane can effectively separate thorium and uranium in rare earths in highly acidic waste liquid, and it is expected to be used in the recovery of rare earths in nuclear waste (Fig. 4b) [75]. Unfortunately, the film material does not show the ability to separate rare earth elements.

Previous reports have shown that ion imprinted polymers are effective adsorbents for selectively adsorbing target ions. A technique without grafting was applied and the target ion Dy (III) was used as the imprinting template by Zheng et al. Subsequently, 6 types of 2,2,6,6-tetramethylpiperidine-1-oxyl (TEMPO)-mediated cellulose (CNCs) imprinted membranes were successfully fabricated by evaporative self-

assembly [77]. These results can also confirm the great potential of the imprinted membrane for the separation of single rare earth elements. Cui et al. fabricated Gd(III) embossed carbon nanotubes (GICNTs) by a simple bionic method to realize ion embossing of membrane materials. Using this method to prepare membrane materials has simple operation and good reproducibility, not only overcomes the difficulty of easy agglomeration and recovery of traditional ion imprinted polymers, but also retains the advantages of high ion selectivity. Through adsorption and permeation experiments, it was found that the membrane achieved selectivity of 2.91 (La/Gd) and 2.49 (Eu/Gd) and exhibited good stability, which should enable the separation and purification of multiple rare earth ions (Fig. 4c) [76]. Therefore, we believe that the combined use of multiple separation materials should become the most promising method for rare earth separation.

It is worth mentioning that the separation of heavy rare earth ions is very difficult due to the extremely low content in the raw ore and the poor dissolution rate, but the membrane separation method is more suitable for the separation of heavy rare earth ions. In view of the current progress in membrane separation technology, our future research should focus on solving the swelling problem of membrane materials and improving stability and recycling performance. Although

membrane separation technology is still rarely used in actual separation, the selective separation ability of membrane technology for REEs has been proven to be effective [78,79].

2.5. Adsorption method

In recent years, solvent extraction has been the preferred method for recovering rare earth resources from waste liquids [80]. However, due to the use of large amounts of organic solvents, which are flammable and toxic, it is not recommended to use them in industrial wastewater with low concentrations of rare earths. In the adsorption method, the target metal ions are applied to the surface of the adsorbent or distributed in the pores of the adsorption carrier by the interaction between the adsorbent and the adsorbate [15]. This not only greatly reduces the use of solvents and additives but also has a wide concentration range, making it a simpler and more environmentally friendly choice [81]. Under the current strict environmental protection regulations, the adsorption method has greater commercial value than solvent extraction technology [82].

The solid phase support used in the adsorbent is selected to have a larger specific surface area, the material can be easily modified with functional groups, the structure is stable, and the repeated use rate is high. Early research focused on the use of traditional porous materials, such as bare silica gel, zeolite and activated carbon. The difference in the binding force between the rare earth ion and the negative charge in the zeolite framework is used for the selective separation of the rare earth ion by the adsorbent material [83,84]. These traditional materials are inexpensive and can improve the adsorption capacity and separation factor by modifying functional groups, compounding with other materials, or changing the structure [85]. Compared with traditional adsorbents, adsorbents made of nanomaterials increase the specific surface area of the material and contribute to the adsorption capacity of the adsorbent. To date, researchers have synthesized and reported a variety of porous materials functionalized with functional groups and new molecularly imprinted materials. It can be used as an aqueous rare earth adsorbent for the purification and separation of mineral leachate and rare earth-containing water resources.

2.5.1. Bio-adsorbents

The development of low-cost and environmentally friendly rare earth separation technology is of great significance for the improvement of traditional separation technology. Adsorbent with natural resources as raw material, low cost, non-toxic, and easily degradable are eco-friendly separation materials [77,86]. In recent years, bio-adsorbents have been studied by many research groups, including natural adsorbent materials such as cellulose, starch, gum arabic, zein, sargassum, brown marine organisms, and bacteria [87,88]. The surface of these natural adsorbents contains polysaccharides and other substances that can be used as binding sites (such as carboxyl groups) to absorb rare earth functional groups and selectively bind metal ions [89]. Since it is extremely difficult to separate adjacent rare earth elements, chemical modification of the binding sites on the surface of biological cells can also enhance the selectivity of bio-adsorbent materials [90].

Currently, some scientists have reviewed the research results of adsorption by bacteria in recent years and discussed in detail how to take advantage of microorganisms to adsorb rare earths and the direction of future research [91]. To better utilize biological materials as rare earth element adsorbents for separation and environmental applications, biomass adsorbents are often used in combination with other substrates. This part of the content is reported later in this article along with other matrix adsorption materials. Although bio-adsorbent materials have many advantages, it is still a problem of how to complete the pre-concentration of adsorbents that we need to solve in order to use bio-adsorbents for industrial scale.

2.5.2. Silica-based sorbents

Adsorption is a surface process, and a larger specific surface area can improve the adsorption performance of the adsorbent. Silicon-based mesoporous materials have extremely high specific surface area and porosity, greatly improving the adsorption capacity and contact efficiency, and the surface of silicon-based materials more easily grafts or copolymerizes functional groups, which plays an important role in improving the selectivity of adsorbents [92,93]. However, silicon-based materials have poorer selectivity to metals than metal adsorbents. This is because rare earth ions are typical Lewis acid ions. Functionalized adsorption materials with groups containing Lewis bases (N, P, -COO, etc.) may be advantageous for rare earth separation [92].

In previous reports, highly efficient adsorption materials were developed by linking polyacrylamide (PAM) with polysaccharides [94]. Later, some researchers used the radial in situ graft copolymerization method to graft-modify the natural polysaccharide gum arabic GA and produce a GA-g-PAM nanocomposite. Compared with the raw materials GA and GA-g-PAM, the specific surface area of GA-g-PAM/SiO₂ is significantly increased (273.55 m²/g), and the desorption reuse rate is as high as three cycles; GA-g-PAM/SiO₂ could be used for the adsorption and recovery of rare earths in aqueous solutions [95]. Iftikhar et al. also investigated and synthesized two different acid-modified cellulose-based silica (CLx/SiO₂) nanocomposites. The experimental results show that the CLx/SiO₂ composite prepared by sulfuric acid modification is more conducive to adsorption. This material has good adsorption properties for Eu(III), La(III) and Sc(III), and can also be used for rare earth element enrichment [43]. However, the above materials do not show selective adsorption between rare earth elements.

To further improve its adsorption efficiency, Fryxell et al. grafted acetamide phosphonic acid (AcPhos), propionamide phosphonic acid (Prop Phos) and other acid-functionalized SAMMS materials onto the surface of mesoporous silica to improve the adsorption performance of the material for lanthanide elements [96]. Gao et al. chemically grafted an ordinary inorganic phosphite phosphorus oxychloride (POCl₃) onto the surface of mesoporous SBA-15 and obtained a new type of phosphoric acid-functionalized mesoporous SBA-15 (P-SBA-15). This material reached the maximum adsorption capacity for Gd(III) within 2 min, and after three repeated adsorptions, the adsorption efficiency still reached 92.3% after repeated adsorption (Fig. 5a) [97]. However, the adsorption performance of these materials under acidic conditions is not ideal. Zhang et al. used diamidophosphorus-functionalized silicon-based materials to achieve efficient separation of Th and Ln under strongly acidic conditions, with a separation factor (SF) of up to 1000 Th/Ln [98]. Unfortunately, this material does not show a highly selective separation between rare earth elements.

The pores of KIT-6 silica material have a high degree of interoperability, which is beneficial to the separation and diffusion of liquids and reduces the risk of pore clogging. It is a promising solid support material for metal ion adsorption. In 2018, Ravi and his colleagues obtained mesoporous silica with accessible adsorption sites and good stability by grafting ethylenediaminetetraacetic acid (EDTA) onto two representative mesoporous silica (KIT-6 and KCC-1) (as shown in Fig. 5b). The results show that it has a large adsorption capacity for Nd³⁺, and the maximum adsorption capacity can reach 109.8 and 96.5 mg/g, respectively. This material can be reused for at least 5 cycles, showing high regeneration stability [99]. Salih et al. grafted amino-rich diethylene triamine (DETA) onto glycidyl methacrylate coated silica SiO₂ particles and synthesis of amino-rich silicon-based composites to improve the adsorption performance of Nd(III) and Gd(III) in water [92]. Florek et al. modified the mesoporous KIT-6 silica material with diglycolamide (DGA)-type ligands and successfully achieved highly selective separation of lanthanide rare earth elements, with the prepared material exhibiting higher selectivity than the commercially available DGA resin [102]. In the follow-up work of Florek and his colleagues, they further synthesized three identical DGA-based hybrid silica adsorbents, applied them to rare earth extraction and purification, and discussed the

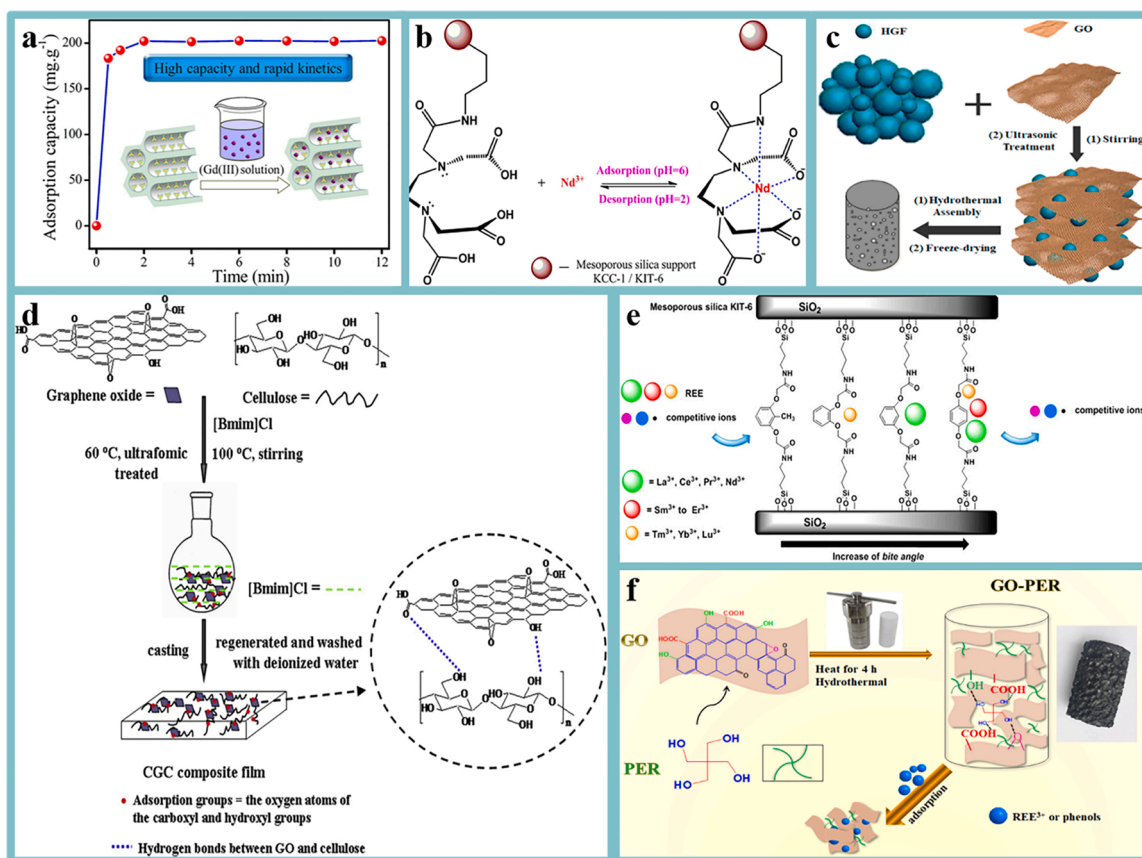


Fig. 5. (a) Inorganic phosphonic acid-modified mesoporous SBA-15 adsorption of Gd(III) [97]. (b) Adsorption-desorption mechanism of Nd^{3+} over KCC-1-EDTA and/or KIT-6-EDTA [99]. (c) Synthetic schematic diagram of HGF-GOx:y composite [100]. (d) Preparation route of CGC adsorbent (inset indicates the interaction between cellulose and GO) [6]. (e) Schematic diagram of the size selective separation of rare earth ions by mesoporous silicon [93]. (f) Preparation of 3D-GO-PER composite material and its application in the adsorption of REE^{3+} and phenols [101].

influence of the chelate angle of the modified functional groups on the selective adsorption of the adsorbents and rare earths [103]. On this basis, Hu et al. grafted a series of tetradentate phenyldioxidamine (PDDA) ligands onto the surface of KIT-6 silica, and selectively separated rare earth ions by adjusting the occlusal angle of the chelating ligand (as shown in Fig. 5e). This material shows obvious size selectivity for rare earths, and selective recovery of rare earths in actual separated samples (bauxite residues) with good reusability [93]. Compared with a single silicon-based adsorbent, different functional groups are grafted onto the surface of silicon-based materials, and the difference in the chelating angle of the functional groups has an important influence on the interaction between the adsorbent and rare earth elements.

Silicon-based materials have poor stability under acidic conditions, and the application of rare earth element separation and recovery in acidic wastewater is limited [104]. In contrast, carbon-based materials have good stability in acidic environments [22]. Ramasamy et al. used two different types of 1-(2-Pyridylazo) 2-naphthol (PAN) grafting techniques. Activated carbon (AC) and silica composite materials were prepared by adapting the hybridization and ligand grafting methods, and the effects of single materials (AC-P, SiLP) and hybrid materials (AC-SiLP) on rare earth removal efficiency were studied [105]. The results show that the hybridization of carbon-based and silicon-based materials can not only retain the adsorption properties of silicon-based materials, but also improve the stability of the materials under acidic conditions due to the addition of carbon-based materials. Therefore, we speculate that the combination of silicon-based materials with materials that are stable in acidic systems, and the grafting of a chelating agent group selective to rare earth elements on the surface of this composite material can effectively improve the absorption and separation

performance of rare earths by silicon-based materials.

2.5.3. Carbon-based adsorption materials

Carbon-based adsorbents based on mesoporous carbon, nanotubes (CNTs), and graphene have been widely used in the separation of rare earth elements [104]. Wang et al. used wrinkled mesoporous silica (WMS) as a template to simulate the pore structure with unique tapered wrinkled pores on carbon materials. They obtained wrinkled mesoporous carbon (WMC), which was further treated with acid to functionalize the oxygen-containing groups on the surface of the pleated mesoporous carbon. The resulting material not only shows great adsorption capacity but also has high selectivity for thorium (Th), which can be used to purify rare earth elements in minerals [22]. This method provides us with ideas for studying new materials. Carbon materials prepared with mesoporous silica as a template not only provide silica pore structures, but also increase the adsorption activity of the material after further acid treatment. Sites are conducive to the adsorption of metal ions.

Since the discovery of graphene, graphene and its derivatives have flourished in various fields due to their excellent physical and chemical properties [106]. The rich oxygen-containing functional groups on the surface of GO show a strong affinity for rare earth elements and a relatively high adsorption capacity, which has great potential as an adsorption material [107]. However, GO materials tend to agglomerate and are not suitable to exist as a single adsorbent. Usually, it is necessary to functionally modify graphene oxide or introduce other matrices to improve its adsorption and separation capacity [108]. In addition, having nitrogen-containing groups on the surface can increase the selectivity to rare earth ions.

In recent years, graphene-based adsorbents have become more selective for rare earth metal adsorption by combining macrocyclic ligands and chelating ligands on the surface of graphene [109]. Zhao et al. coated the surface of GO by one-step ultrasonic treatment with tris (4-aminophenyl) amine (TAPA), which is rich in amino and phenyl groups, to synthesize a three-dimensional (3D) graphene oxide-tris (4-aminophenyl) amine (GO-TAPA x:y) composite material. The porous structure of the composite material provides a large specific surface area for the adsorption of rare earth ions. The newly added lone electron of the amino group on TAPA has a different affinity with rare earth ions, which increases the adsorption selectivity for a single rare earth element, and this material shows high stability and reusability in actual water samples [110]. The coating on the surface of GO partially improves the shortcomings of the material itself, which easily agglomerates, and the chelating ligand increases the selective adsorption of the material to rare earth ions. Perhaps it is not necessary to chemically graft the chelating agent group to modify the surface of the adsorbent material, and it is also feasible to coat the surface of the adsorbent with amino lone electrons.

According to research results, ionic liquids can solve the problem of poor solubility of cellulose in water and common organic solvents [111]. Then, Yan et al. used the solubility of cellulose in ionic liquids (RTILs) to disperse cellulose in GO at room temperature to prepare GO/cellulose composite membranes (CGCs) (as shown in Fig. 5d). The adsorption evaluation using the rare earth element Ce as the simulation object confirmed that the GO/cellulose composite material has good adsorption performance for rare earth ions and the fabricated CGC composite has high acid resistance. It is expected to be an effective adsorbent for the enrichment and separation of rare earth elements in wastewater [6]. It is a pity that CGC does not show the selective separation ability of rare earth elements.

Unlike two-dimensional graphene, three-dimensional graphene exhibits unique amphiphilicity and a larger specific surface area due to lamination and porosity in addition to the excellent properties of two-dimensional graphene itself. Liao et al. applied a hydrothermal self-assembly method and used the interaction of intermolecular hydrogen bonds to successfully prepare three-dimensional graphene oxide pentaerythritol (3D-GO-PER) (Fig. 5f), which is rich in oxygen-containing functional groups and provides more active sites for the adsorbent. Through indirect adsorption experiments, the results prove that the composite material has great potential for phenolic compounds and rare earth elements and has great potential for improving the pollution of the water system [101]. Compared with the two-dimensional graphene structure, the three-dimensional interlayer porosity of the material is conducive to the increase of the adsorption capacity. However, a large amount of rare earth ions may remain in the material pores during the desorption and regeneration of the adsorbent, and the adsorption performance may be reduced. We believe that the recycling rate of this adsorbent material is still to be discussed.

The GO composite of biomass material has good adsorption, low toxicity and reusability, and provides more active sites for the chelation of metal ions [109]. Modification of GO by chemical reagents has been reported in the literature. However, this could lead to secondary pollution and the reduction of many adsorption sites due to the formation of amide and ester bonds. Researchers have synthesized a three-dimensional graphene oxide-based porous composite material (3D-GO-CZ) composite material by a simple hydrothermal treatment method, using natural material zein (CZ) with good adsorption properties as a crosslinking agent. This material can reach the equilibrium value of the adsorption of rare earth ions within 200 min, but unfortunately, the article does not indicate the selective separation performance of this material for rare earth ions [112]. In 2018, Xu et al. used a green and simple hydrothermal method and selected high-gluten flour (HGF) as the binder and support for the fixation of graphene oxide (GO) to prepare a loose-porous 3D HGF-GO composite material (as shown in Fig. 5c). This composite material has a relatively stable network

structure and provides many metal ion-chelations of the active sites. It has not only good rare earth adsorption performance but also has good reusability. And the HGF-based adsorbents are also more environmentally friendly and biodegradable [100]. Later, they used egg white (EW) with abundant cross-linking sites as a structural support and use a take one-step hydrothermal method to embed the EW into GO to produce a composite material (3D-CEG) with a porous three-dimensional structure. This material can also maintain a high removal rate of rare earths (La, Y, Yb, Er and Nd) in actual water bodies and can maintain 100% even when repeated five times at a lower concentration [113]. The combination of different materials can not only maintain the adsorption performance of the raw materials, but also produce interesting new features. The combination of biomass materials and carbon-based materials reduces the use of chemical reagents and makes the adsorption materials biodegradable. However, there are few studies on biomass materials with selective adsorption of rare earth elements, and it is still need to be explored further.

2.5.4. Coordination polymers

Coordination polymers are a new type of ordered porous crystalline material, including metal organic frameworks (MOFs) and covalent organic frameworks (COFs). It has been reported to be used for the adsorption of rare earth elements [114,115]. As shown in Table 2, we summarize the selective separation of rare earth elements for some coordination polymers. Metal organic frameworks (MOFs) are porous crystalline materials linked by metal ions or clusters and organic ligands. They have a large specific surface area, unsaturated metal sites, and tunable pore size. They have great potential as rare earth element adsorption materials [116–118]. Ahmed et al. synthesized ZIF-8 nanoparticles with different pore diameters using a green template-free method, which exhibited good adsorption properties for La^{3+} , Sm^{3+} , and Dy^{3+} in aqueous solutions and showed higher selective adsorption for heavy rare earth ions [114]. This kind of nanomaterial synthesized by a template-free method is very interesting. This method avoids the loss of ions as a template, and may be widely used for pore size control of other kinds of adsorbents.

To improve the selective separation ability of the coordination polymer (CP), it is necessary to modify the coordination polymer. Xu et al. modified it with an oxygen-containing group 4,4'-azoxydibenzoic acid and utilized the electrostatic interaction and coordination effect between rare earth ions and oxygen-containing groups to efficiently and selectively capture rare earth target ions of lanthanides [119]. Kavun et al. functionally modified MIL-101(Cr) with organophosphorus compounds (TBP, HDEHP and Cyanex-272) [120]. Different oxygen-containing groups and rare earth ions on the surface of the adsorbent produce different complexing effects, which makes the modified material of this kind of work show high selectivity for rare earth ions. When a Lewis base with a higher affinity for rare earth ions is used as a complexing ligand to modify MOF materials, it is expected to be a stable and highly selective material in an acidic environment [119, 121]. Choose MIL – 101(Cr) as the support has a significantly stable structure in acidic environments. Decker et al. functionalized MIL – 101 (Cr) with carbamoyl phosphine oxide (CMPO), which can be effectively used to selectively separate uranium in oxalate mixtures [122]. Jeroen et al. used formyl methyl phosphine oxide (CMPO) to functionalize MIL-101(Cr). This material is stable under acidic conditions and not only has the maximum adsorption capacity of Eu up to 12.45 mg/g, but also the separation factor for Eu and Y can reach 3.2 [121]. It is very common to graft functional groups on the adsorption surface to obtain materials with high adsorption capacity or high selectivity, and it is becoming increasingly difficult to innovate materials.

Combining MOF materials with resins, cellulose and other materials to construct nanocomposites is also expected to improve the stability of MOFs in aqueous environments. In addition, some new physical and chemical properties may appear after combining the individual materials. Therefore, research on composite materials is necessary for

Table 2
Separation of rare earth elements by different coordination polymers.

Material	Target ion	Maximum adsorption capacity (mg·g ⁻¹)	Optimal experimental conditions	Separation factor	References
ZIF-8 NPs	La	28.8	t = 7 h, pH = 7, C ₀ = (12–200 mg/L), T = room temperature.	Dy/La = 50.76	[114]
	Sm	281.1		Dy/Sm = 2.91	
	Dy	430.4		–	
Fe ₃ O ₄ @ZIF-8	UO ₂	523.5	t = 2 h, pH = 3, C ₀ = (5–350 mg/L)	–	[134]
CMPO@MIL-101 (Cr)	Eu	12.5	t = 24 h, pH = 5.5, C ₀ = (10–500 ppm), T = 21 °C.	Eu/Y = 3.2	[120]
MIL-101-C50	Nd	27.35		Eu/Zn = 8.5	
	Gd	31.17		Er/Nd = 9.5	
	Er	48.91		Er/Gd = 4.5	
MIL-101-H50	Nd	34.86		Er/Nd = 22.8	
	Gd	44.91		Er/Gd = 7.7	
	Er	57.47		–	
MIL-101-T50	Nd	0.39		–	
	Gd	0		Er/Gd = 17.5	
	Er	34.90		–	
EDTA-CS@ZIF-8	La	256.4	t = 24 h, pH = 6, T = 298 K, C ₀ = (5–45 mg·L ⁻¹), Adsorbent dose = 1.5 mg/L	–	[128]
	Eu	270.3		–	
	Yb	294.1		–	
[NH ₄] ⁺ [COF-SO ₃ ⁻]	Th	395	m (adsorbent) = 10 mg, V = 20 mL, t = 120 min, pH = 2.8.	–	[130]
	U	208		Th/U = 9.4	
	Eu	189		ThEu = 9.4	
GA-g-PAM/SiO ₂	Ce	181	Adsorbent dose = 3.5 mg/L, pH = 6, T = 298k	Th/Ce = 10.3	[95]
	Eu	10.11		–	
	La	7.9		–	
P-SBA-15	Nd	12.24	Adsorbent dose = 0.5 mg/L, t = 10 min, T = 303 k, C ₀ = (50 ~ 200 mg·L ⁻¹).	–	[97]
	Sc	11.05		–	
	Gd	202.92		–	
SBA-15-O-DMAP	Th	–	Adsorbent dose = 4 mg/L, pH = 3.8, t = 2 h, T = 298 K, C ₀ = 100 mg·L ⁻¹ .	Th/Ln > 1000	[98]
	U	–		Th/U > 50	
	Ln	–		–	

industrial applications [123]. Currently, many composites of graphene oxide and metal-organic frameworks (MOFs) are known to be used in fluorescent probes [124], anti-corrosion coatings [125], lithium electronic batteries and supercapacitors [126], among others. The mechanical properties of MOF materials are poor. In order to improve their mechanical properties, they can be combined with a stable matrix material [127]. Feng et al. used ZIF-8 and chitosan as raw materials to prepare a new type of adsorption material. Since the combined adsorption sites of the composite material were reduced, they further introduced an EDTA chelating agent to compensate for some adsorption active sites. This material showed a large adsorption capacity for La(III), Eu(III) and Yb(III) [128]. In the above research, although the combination of different materials has the advantages of the two materials, the composite materials cover each other, and the active adsorption sites may be reduced. The introduction of chelating group modification again compensates for the active sites, but it also increases the difficulty of the material preparation process. In 2020, Wu et al. successfully synthesized Zn-BTC MOF/nanoporous graphene composites by a simple local combustion strategy, which combines the advantages of graphene and metal-organic framework materials in the separation of a single rare earth element [129]. This material does not require the help of organic solvents and additives. It is not only environmentally friendly but also structurally stable and far superior to the reported materials for selective separation of single rare earth elements. Among them, the separation factor of Ce/Lu is as high as 12,081.04, and it is expected to be used in industrial production. Unfortunately, the recycling rate of materials has not been mentioned.

Following the demonstration of the effectiveness of metal-organic framework (MOF) materials in adsorption separation, the potential of covalent organic framework (COF) as a porous ordered crystalline material in rare earth separation was also discovered. Xiong et al. applied COF to separate rare earths, thorium, and uranium [130]. They found that COF has high adsorption capacity and can meet the industrial

separation standards of thorium (as shown in Fig. 6). As a new material, COF has great potential in the field of adsorption, but whether the COF material has the ability to selectively separate rare earth elements remain to be verified.

2.5.5. Magnetic nano adsorbents

Although adsorbents have been widely used for rare earth adsorption and separation in recent years, they inevitably have limitations, such as poor stability and the possibility of secondary contamination. The introduction of magnetic nanoparticles solves this very problem, and the addition of magnetic substances is also conducive to the control of rare earth separation, recovery and reuse [131]. At present, functionalized magnetic nanomaterials have been shown to be the most promising sustainable adsorbents for the selective adsorption of rare earth ions from aqueous solutions. Dupont et al. reported the use of N-[(3-trimethoxysilyl) propyl] ethylenediaminetriacetic acid (TMS-EDTA) to modify the surface of magnetic (Fe₃O₄) and non-magnetic (SiO₂ and TiO₂) nanoparticles [132]. Almeida et al. reported the results of superparamagnetic Fe₃O₄ nanoparticles functionalized with DTPA (diethylene triamine pentaacetic acid) [133]. The experimental results show that the magnetic nano-adsorbent material modified with the chelating agent has higher adsorption capacity and selective adsorption. Although the magnetic adsorption material has good reusability, its adsorption capacity and selectivity for rare earth ions in aqueous solutions are not ideal. Therefore, how to combine the recyclability of the functionalized magnetic nanomaterials with the high adsorption capacity and high selectivity of the non-magnetic nanomaterials to improve the structural stability of the magnetic nanoparticles will continue to be the focus of research in the field of rare earth element separation.

2.5.6. Gel materials

Among the developed adsorption materials, hydrogels have a stable pore interpenetrating structure and are rich in functional chelating

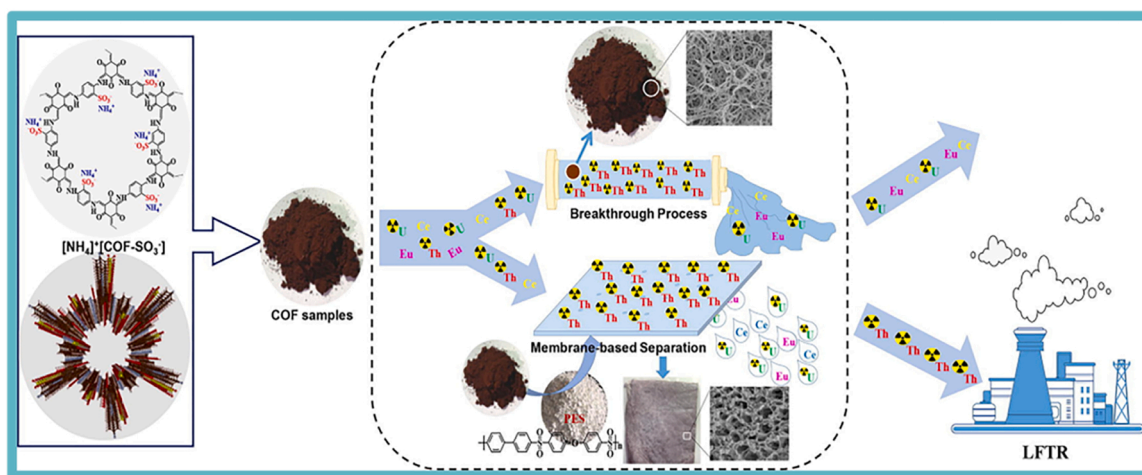


Fig. 6. Preparation and application of adsorbent $[\text{NH}_4]^+[\text{COF-SO}_3^-]$ [130].

groups (-COOH, -SO₃H, -NH₂); hydrogels have different affinities for different metal ions, which is conducive to the adsorption of characteristic ions. Therefore, it is considered as one of the most promising adsorption and separation materials [135]. However, the lack of mechanical strength of this material limits its application. It has been shown that doping with nanomaterials such as silica and graphene has been proven to improve the mechanical strength of this composite material. Almeida et al. fabricated a new type of PAA-S hydrogel nanofibers (PAA-S-HNFs) by applying the colloidal electrospinning method to incorporate silica, and the results showed significant adsorption capacity for rare earth ions (Ln^{3+}) [136]. Although the mechanical strength of the material has been improved, the electrospinning process has increased the difficulty of material preparation and is not suitable for large-scale production. By using the low-temperature polymerization method to introduce GO into the gel, Yang et al. not only improved the mechanical strength of the gel, but also showed that the adsorption efficiency of La^{3+} could reach more than 95% [135]. Moreover, this kind of adsorption material has a simple synthesis process, high efficiency, environmental protection, and sustainability, and has great potential in wastewater recovery and separation of rare earth elements.

In the adsorption method, the contact time between the adsorbent material and the solution, the pH and temperature of the solution, and the concentration of the target ion in the solution will all affect the adsorption capacity of the target ion by the adsorbent [137]. For the calculation of the maximum adsorption capacity of the adsorbent, it is usually necessary to study the adsorption kinetics of the adsorbent, the influence of different pH values, adsorption thermodynamics and adsorption isotherm experiments [138]. Then, under the optimal contact time, pH value and temperature conditions, through the adsorption isotherm of the adsorbent material, the adsorption capacity changes in different concentrations of rare earth solutions are fitted by the Langmuir and Freundlich formulas to obtain the maximum adsorption capacity of the adsorbent [114,139].

The adsorption method is more suitable for solutions with a low content of rare earth ions, through selective enrichment of rare earth ions, to achieve the purpose of separation and recovery of rare earths. The current research on the separation of rare earth elements by adsorption methods has focused on graft modification on the surface of the adsorbent and its composite with other matrix materials. Research on adsorbents with high adsorption capacity and high selectivity is still of great significance.

2.6. Electrochemical process

In addition to the common rare earth separation technologies

mentioned above, new technologies such as electrochemistry have opened up new ideas for rare earth separation due to their low energy consumption and environmental friendliness. Fang et al. proposed a new rare earth separation method, using a series of rare earth complexes containing redox ligands $[\text{RE}(\text{TriNOx})]/[\text{RE}(\text{TriNOx})\text{thf}]$ to achieve electrokinetic separation of binary rare earth mixtures [140]. This method provides a new path for metal separation. Vapnik and his colleagues combined traditional ion exchange resins with redox active groups to synthesize a redox copolymer poly(ferrocene propyl methacrylamide-methacrylic acid) (P(FPMam-co-MAA)) (Fig. 7a) [141]. The polymer not only promotes the selective binding of molecules, but also provides a new idea for electrochemical regeneration of ion exchange copolymers.

Li et al. used electrochemical dialysis technology to separate and extract the target ion Sc^{3+} by using a charged membrane to drive the potential difference, which demonstrated the high selection and low energy consumption of the electrodialysis (ED) system for rare earth separation [142]. However, during the electrodialysis process, some particles and colloidal particles contained in the material may interact with the membrane and deposit on the surface and pores of the membrane, which reduces the reuse rate of the membrane and is not suitable for long-term industrial operation. Gueroul et al., taking neodymium-iron-boron (NdFeB) magnets with high rare earth content as an example, proposed the use of an environmentally friendly multi-stage plasma separation technology to separate and recover high-purity single rare earth elements from NdFeB magnets (Fig. 7b) [143]. The plasma quality filter is still in the pre-development stage, and further evaluation is still needed in the actual sample processing link. It is undeniable that this concept has high potential value, and it is hoped that after further research by researchers, it can be used for large-scale industrial treatment of rare earths.

Electrochemistry provides a new idea for the separation and research of metals. The current electrochemical separation method has been applied to the separation and purification of single rare earth elements due to its environmental protection, low energy consumption, simple operation and continuous operability. However, if the metal needs to be liquid in this process, the electrolyte is required to have a higher melting point than the metal, and the choice of electrolytic cell and electrolyte is limited. Before large-scale use in industrial production, it is extremely important to select suitable electrolytic cell materials and process research and development.

2.7. Separation technology post-processing

Considering environmental issues, the process of disposal of the

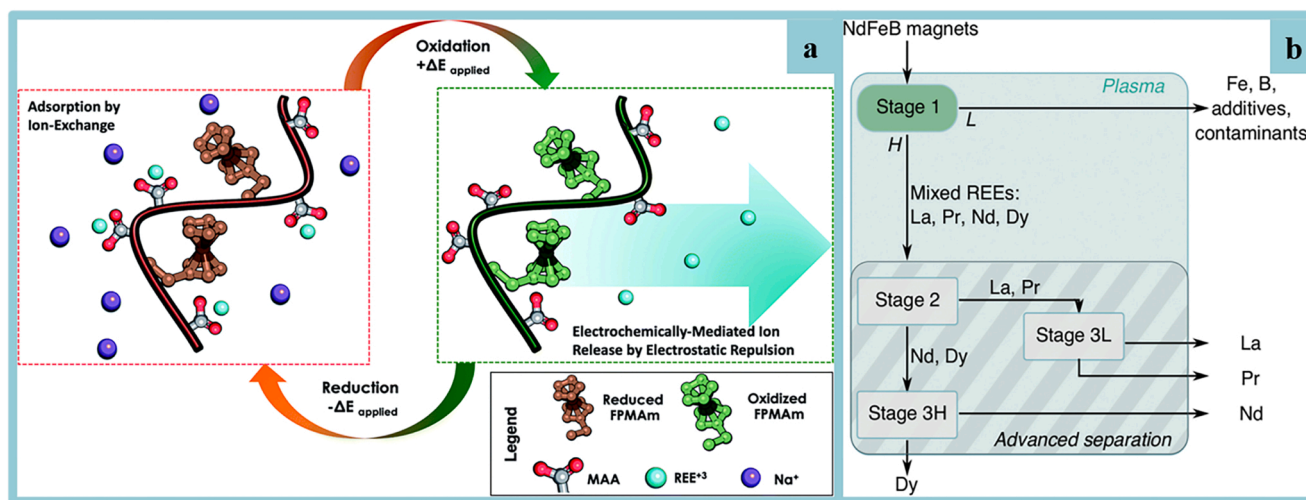


Fig. 7. (a) Schematic diagram of the reversible capture and release of REEs by electrochemical regeneration ion exchange [141]. (b) Schematic diagram of plasma recovery process of NdFeB magnet [143].

generated waste is very important. However, most of the above separation processes are in the experimental stage and have not been put into large-scale industrial production, so the post-treatment of the product is rarely mentioned. Combined with relevant reports, we learned that the post-processing research of resin materials, membrane materials and adsorption materials in the above separation technologies mainly focuses on the recycling and reuse of materials [6,35,41,75,101]. The material is treated with HCl or HNO₃ solution to elute the rare earth elements attached to it, and then used for reuse. The solvent used in the separation technology is recycled through vacuum distillation and other processes to continue to be used in the separation process [55–60,97]. Finally, after the rare earth elements in the solution precipitate, the precipitate is sintered to obtain rare earth oxide, and the waste gas is discharged into the air after a simple treatment [25–27]. Therefore, the post-processing after separation is still a problem that needs our attention.

2.8. Conclusion and prospects

Since the 1990 s, rare earths have experienced rapid development in the fields of inorganic materials, petrochemicals, and electronics. Due to their relatively similar ionic radius, rare earths are usually found in multi-element mixtures. However, industrial applications place higher demands on the purity of single rare earths. Therefore, in recent years, researchers have focused on developing separation and purification technology for rare earths. This article reviews the recent development status and limitations of rare earth element separation technologies.

Among the commonly used separation technologies, the traditional stepwise method to obtain a single rare earth element requires multiple dissolution-crystallization operations. Considering the production cycle and cost, it is currently rarely used in the separation of a single rare earth element. It is worth noting that the selective crystallization technology of the metal organic framework not only reduces the use of additives in the stepwise method, but also has a higher selectivity to a single rare earth element. However, the current research on this topic is still in its infancy, and whether it can be used in industrial production remains to be investigated. The purity of the single rare earth element obtained by the ion exchange method is extremely high, but the continuous processing capacity is limited, the production cycle is long, and in view of the poor stability of the ion exchange resin material, the resin material must be replaced regularly to obtain the high purity single rare earth element, resulting in high process costs. At present, ion exchange technology has almost been replaced by solvent extraction, but to obtain high purity rare earth elements, ion exchange can be the preferred

technology. Solvent extraction is currently the most commercialized technology in the field of rare earth separation. Because of its large processing capacity, continuous production and remarkable separation effect, it can be recommended for industrial production. However, it also has negligible shortcomings, consumes many organic extractants, and has difficulty regenerating the organic phase after stripping. Considering economic reasons, it is not recommended to treat low-concentration rare earth solutions. As an emerging technology, the membrane separation method has significant advantages of low energy consumption and high efficiency, but it still does not meet the requirements of industrial production. The regeneration performance of the membrane needs to be further improved, and the required equipment needs to be developed. The adsorption method is suitable for the treatment of low-concentration rare earth solutions due to its enrichment effect on the rare earth ions in the solution. Although the adsorption method has many advantages, it is difficult for the adsorption material to have both high adsorption capacity and high selectivity at the same time, and the current research is still in the experimental development stage and has not been tested by large-scale production. Electrochemical separation of metal elements has the advantages of continuous operation, environmental protection, low energy consumption, and simple operation. It is suitable for large-scale industrial extraction of rare earth elements in minerals or compounds. However, there are still few studies on the separation of a single rare earth element, and a large amount of experimental data is needed to support it. In general, there is still great potential for further improving rare earth separation technology and increasing separation efficiency.

In the future, the development of green and high-quality rare earth separation technology should focus on the following aspects: (a) Solve the environmental problems caused by traditional separation technology, supplement and improve the rare earth separation industry specifications, and establish a green process supervision system; (b) Adhere to the orientation of green development and develop green separation technology with higher selectivity, stability and cost-effectiveness; (c) Together with other separation technologies to establish a new combined separation system with complementary advantages; (d) According to different scenarios, select the appropriate separation technology; (e) Based on the long-term operation and development of the industry, further develop the process technology with low energy consumption, high efficiency, low cost and large processing capacity; (f) Taking into account the protection of the environment and ecology and the recycling of resources, it is of great significance to the recovery and treatment of secondary resources containing rare earths.

CRediT authorship contribution statement

Ziying Chen: Investigation, Writing – original draft, Editing. **Zhan Li:** Writing – review & editing. **Jia Chen:** Writing – review & editing. **Parashuram Kallem:** Writing – review & editing. **Fawzi Banat:** Writing – review & editing. **Hongdeng Qiu:** Writing – review & editing, Project administration, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by National Key Research and Development Program of China (2019YFD1002403), the National Natural Science Foundation of China (Nos. 22050410282, 21822407, 21974146 and 22074154), Youth Innovation Promotion Association CAS (2021420) and the Foundation for Sci & Tech Research Project of Gansu Province (20JR10RA045 and 20JR5RA573).

References

- [1] A. Mehdi, T. Rezvan, T. Meisam, Extraction and separation of rare earth elements by adsorption approaches: current status and future trends, *Sep. Purif. Rev.* 50 (2020) 417–444.
- [2] J. Lie, J. Liu, Selective recovery of rare earth elements (REEs) from spent NiMH batteries by two-stage acid leaching, *J. Environ. Chem. Eng.* 9 (2021), 106084.
- [3] Q. Yan, C. Liu, X. Zhang, L. Lei, C. Xiao, Selective dissolution and separation of rare earths using guanidine-based deep eutectic solvents, *ACS Sustain. Chem. Eng.* 9 (2021) 8507–8514.
- [4] C. Peng, B. Xue, W. Wenyuan, L. Baokuan, A sustainable green technology for separation and simultaneous recovery of rare earth elements and fluorine in bastnaesite concentrates, *Sep. Purif. Technol.* 274 (2021), 118380.
- [5] A.T. Lima, L. Ottosen, Recovering rare earth elements from contaminated soils: critical overview of current remediation technologies, *Chemosphere* 265 (2021), 129163.
- [6] Y. Hao, Y. Cui, J. Peng, N. Zhao, S. Li, M. Zhai, Preparation of graphene oxide/cellulose composites in ionic liquid for Ce (III) removal, *Carbohydr. Polym.* 208 (2019) 269–275.
- [7] B.R. Reddy, J.R. Kumar, Rare earths extraction, separation, and recovery from phosphoric acid media, *Solvent Extr. Ion.-Exc.* 34 (2016) 226–240.
- [8] W. Gwenzi, L. Mangori, C. Danha, N. Chaukura, N. Dunjana, E. Sanganyado, Sources, behaviour, and environmental and human health risks of high-technology rare earth elements as emerging contaminants, *Sci. Total Environ.* 636 (2018) 299–313.
- [9] N. Swain, S. Mishra, A review on the recovery and separation of rare earths and transition metals from secondary resources, *J. Clean. Prod.* 220 (2019) 884–898.
- [10] D. Tanushree, K. Ki-Hyun, U. Minori, E.K. Eilham, J. Byong-Hun, D. Akash, Y. Seong-Taek, Global demand for rare earth resources and strategies for green mining, *Environ. Res.* 150 (2016) 182–190.
- [11] Y. Bian, S. Guo, L. Jiang, J. Liu, K. Tang, W. Ding, Recovery of rare earth elements from NdFeB magnet by VIM-HMS method, *ACS Sustain. Chem. Eng.* 4 (2015) 810–818.
- [12] P. Yang, Q. Zhuang, Y. Li, J. Gu, Green separation of rare earth elements by valence-selective crystallization of MOFs, *Chem. Commun.* 55 (2019) 14902–14905.
- [13] M. Hermassi, M. Granados, C. Valderrama, C. Ayora, J.L. Cortina, Recovery of rare earth elements from acidic mine waters by integration of a selective chelating ion-exchanger and a solvent impregnated resin, *J. Environ. Chem. Eng.* 9 (2021), 105906.
- [14] Y. Zhou, X. Xue, H. Yang, S. Song, X. Huang, Novel harmless utilization of bayan obo tailings: separation and recovery of iron and rare earth, *Ind. Eng. Chem. Res.* 59 (2020) 13682–13695.
- [15] Y. He, S. Guo, K. Chen, S. Li, L. Zhang, S. Yin, Sustainable green production: a review of recent development on rare earths extraction and separation using microreactors, *ACS Sustain. Chem. Eng.* 7 (2019) 17616–17626.
- [16] R.G. Silva, C.A. Morais, É.D. Oliveira, Selective cerium removal by thermal treatment of mixed rare earth oxalates or carbonates obtained from non-purified rare earth sulphate liquor, *Miner. Eng.* 139 (2019), 105865.
- [17] L. J. R. M. G. O, L.C. J. Integration of nanofiltration membranes in recovery options of rare earth elements from acidic mine waters, *J. Clean. Prod.* 210 (2019) 1249–1260.
- [18] B.E. Cole, I.B. Falcones, T. Cheisson, B.C. Manor, P.J. Carroll, E.J. Schelter, A molecular basis to rare earth separations for recycling: tuning the TrINOx ligand properties for improved performance, *Chem. Commun.* 54 (2018) 10276–10279.
- [19] T. Liu, J. Chen, Extraction and separation of heavy rare earth elements: a review, *Sep. Purif. Technol.* 276 (2021), 119263.
- [20] F. Xie, T.A. Zhang, D. Dreisinger, F. Doyle, A critical review on solvent extraction of rare earths from aqueous solutions, *Miner. Eng.* 56 (2014) 10–28.
- [21] J. Chen, M. Meng, H. Dong, Y. Wu, Y. Yan, C. Li, L. Chen, An overview on membrane strategies for rare earths extraction and separation, *Sep. Purif. Technol.* 197 (2018) 70–85.
- [22] Z. Wang, A.T. Brown, K. Tan, Y.J. Chabal, K.J. Balkus, Selective extraction of thorium from rare earth elements using wrinkled mesoporous carbon, *J. Am. Chem. Soc.* 140 (2018) 14735–14739.
- [23] H. Onoda, A. Iinuma, Selective preparation of neodymium phosphates from iron mixed solution by two-step precipitation, *J. Environ. Chem. Eng.* 8 (2020), 104083.
- [24] E.O. Opare, E. Struhs, A. Mirkouei, A comparative state-of-technology review and future directions for rare earth element separation, *Renew. Sust. Energ. Rev.* 143 (2021), 110917.
- [25] C. Liu, Q. Yan, X. Zhang, L. Lei, C. Xiao, A one-pot process based on P44414Cl-HCl aqueous biphasic system for recovering rare earth elements from NdFeB permanent magnet, *Chin. Chem. Lett.* (2021), <https://doi.org/10.1016/j.ccl.2021.07.026>.
- [26] Y. Lu, H. Wei, Z. Zhang, Y. Li, G. Wu, W. Liao, Selective extraction and separation of thorium from rare earths by a phosphorodiamidate extractant, *Hydrometallurgy* 163 (2016) 192–197.
- [27] D. Prodius, M. Klocke, V. Smetana, T. Alammari, M.P. Garcia, T.L. Windus, I. C. Nlebedim, A. Mudring, Rationally designed rare earth separation by selective oxalate solubilization, *Chem. Commun.* 56 (2020) 11386–11389.
- [28] B.E. Cole, T. Cheisson, J.J.M. Nelson, R.F. Higgins, M.R. Gau, P.J. Carroll, E. J. Schelter, Understanding molecular factors that determine performance in the rare earth (TrINOx) separations system, *ACS Sustain. Chem. Eng.* 8 (2020) 14786–14794.
- [29] N. Sui, K. Huang, Separation of rare earths using solvent extraction consisting of three phases, *Hydrometallurgy* 188 (2019) 112–122.
- [30] Zhao Xiang, Wong Matthew, Mao Chengyu, Trieu X. Thuong, Zhang Jian, Feng Pingyun, Bu Xianhui, Size-selective crystallization of homochiral camphorate metal-organic frameworks for lanthanide separation, *J. Am. Chem. Soc.* 136 (2014) 12572–12575.
- [31] X. Yin, Y. Wang, X. Bai, Y. Wang, L. Chen, C. Xiao, J. Diwu, S. Du, Z. Chai, T. E. Albrecht-Schmitt, S. Wang, Rare earth separations by selective borate crystallization, *Nat. Commun.* 8 (2017) 14438.
- [32] Y. Wang, H. Lu, X. Dai, T. Duan, X. Bai, Y. Cai, X. Yin, L. Chen, J. Diwu, S. Du, R. Zhou, Z. Chai, T.E. Albrecht-Schmitt, N. Liu, S. Wang, Facile and efficient decontamination of thorium from rare earths based on selective selenite crystallization, *Inorg. Chem.* 57 (2018) 1880–1887.
- [33] M. Seno, Utilization of ion-exchanger in organic syntheses, *J. Chem. Soc.* 25 (1967) 78–88.
- [34] B. Chen, M. He, H. Zhang, Z. Jiang, B. Hu, Chromatographic techniques for rare earth elements analysis, *Phys. Sci. Rev.* 2 (2017), <https://doi.org/10.1515/psr-2016-0057>.
- [35] E.K. Hulet, R.G. Guttmacher, M.S. Coops, Group separation of the actinides from the lanthanides by anion exchange, *J. Inorg. Nucl. Chem.* 17 (1961) 350–360.
- [36] M.A.R. Onal, S. Dewilde, M. Degri, L. Pickering, B. Saje, S. Riaño, A. Walton, K. Binnemans, Recycling of bonded NdFeB permanent magnets using ionic liquids, *Green. Chem.* 22 (2020) 2821–2830.
- [37] E. Yamada, H. Murakami, S. Nishihama, K. Yoshizuka, Separation process of dysprosium and neodymium from waste neodymium magnet, *Sep. Purif. Technol.* 192 (2018) 62–68.
- [38] A. Imteaz, K.A. Keshab, L. Yu-Ri, H.R. Kyung, K. Kyoung-Ku, A. Wha-Seung, Ionic liquid entrapped UiO-66: Efficient adsorbent for Gd3+ capture from water, *Chem. Eng. J.* 370 (2019) 792–799.
- [39] J. Zhang, J. Anawati, Y. Yao, G. Azimi, Aerometallurgical extraction of rare earth elements from a ndfeb magnet utilizing supercritical fluids, *ACS Sustain. Chem. Eng.* 6 (2018) 16713–16725.
- [40] W. Chen, J. Jiang, X. Lan, X. Zhao, H. Mou, T. Mu, A strategy for the dissolution and separation of rare earth oxides by novel Brønsted acidic deep eutectic solvents, *Green. Chem.* 21 (2019) 4748–4756.
- [41] W. Abraham, P. Avraham, Impregnated resins: metal-ion complexing agents incorporated physically in polymeric matrices, *Isr. J. Chem.* 17 (1978) 307–315.
- [42] J. Barrat, G. Bayon, X. Wang, S. Le Goff, M. Rouget, B. Gueguen, D. Ben Salem, A new chemical separation procedure for the determination of rare earth elements and yttrium abundances in carbonates by ICP-MS, *Talanta* 219 (2020), 121244.
- [43] S. Iftikhar, V. Srivastava, M. Sillanpää, Enrichment of lanthanides in aqueous system by cellulose based silica nanocomposite, *Chem. Eng. J.* 320 (2017) 151–159.
- [44] L. Sun-Chol, K. Sok-Chol, K. Chung-Su, K. Chol-Ju, K. Chol-Jin, Separation of samarium, europium and gadolinium in high purity using photochemical reduction-extraction chromatography, *Hydrometallurgy* 178 (2018) 181–187.
- [45] H. Knutson, A. Holmqvist, B. Nilsson, Multi-objective optimization of chromatographic rare earth element separation, *J. Chromatogr. A* 1416 (2015) 57–63.
- [46] R.M. Ashour, M. Samouhos, E. Polido Legaria, M. Svård, J. Höglblom, K. Forsberg, M. Palmiöf, V.G. Kessler, G.A. Seisenbaeva, Å.C. Rasmuson, DTPA-functionalized silica nano- and microparticles for adsorption and chromatographic separation of rare earth elements, *ACS Sustain. Chem. Eng.* 6 (2018) 6889–6900.

- [47] L.K. Sinclair, J.W. Tester, J.F.H. Thompson, R.V. Fox, The role of water in extraction and separation of rare earth elements in supercritical carbon dioxide, *J. Supercrit. Fluids* 136 (2018) 180–188.
- [48] M. Spadina, K. Bohinc, T. Zemb, J. Dufreche, Colloidal model for the prediction of the extraction of rare earths assisted by the acidic extractant, *Langmuir* 35 (2019) 3215–3230.
- [49] S. Ilyas, H. Kim, R. Ranjan Srivastava, Extraction equilibria of cerium(IV) with Cyanex 923 followed by precipitation kinetics of cerium(III) oxalate from sulfate solution, *Sep. Purif. Technol.* 254 (2021), 117634.
- [50] S.F. Douglas, Principles and practices of solvent extraction, in: J. Rydberg, M. Cox, C. Musikas, G.R. Choppin (Eds.), Revised and Expanded, second edition, Marcel Dekker, New York, 2004, p. 760. ISBN 0 8247 5053 2, *J. Chem. Technol. Biot.* 80 (2005) 359–360.
- [51] C. Bie, Y. Gao, J. Su, Y. Dong, X. Guo, X. Sun, The efficient separation of thorium from rare earth using oxamic acid in hydrochloric acid medium, *Sep. Purif. Technol.* 251 (2020), 117358.
- [52] X. Peng, J. Su, H. Li, Y. Cui, J. Yong Lee, G. Sun, Theoretical elucidation of rare earth extraction and separation by diglycolamides from crystal structures and DFT simulations, *J. Rare Earths* 39 (2020) 3858–3865.
- [53] Z. Liu, J. Wu, X. Liu, W. Wang, Z. Li, R. Xu, Y. Ding, J. Wang, Recovery of neodymium, dysprosium, and cobalt from NdFeB magnet leachate using an unsymmetrical dialkylphosphinic acid extractant, INET-3, *J. Rare Earths* 38 (2020) 1114–1118.
- [54] B. Dewulf, N.K. Batchu, K. Binnemans, Enhanced separation of neodymium and dysprosium by nonaqueous solvent extraction from a polyethylene glycol 200 phase using the neutral extractant cyanex 923, *ACS Sustain. Chem. Eng.* 8 (2020) 19032–19039.
- [55] S. Pavón, A. Fortuny, M.T. Coll, A.M. Sastre, Neodymium recovery from NdFeB magnet wastes using Primene 81R-Cyanex 572 IL by solvent extraction, *J. Environ. Manag.* 222 (2018) 359–367.
- [56] Y. Dong, X. Guo, Y. Wang, Z. Zhao, C. Huang, X. Sun, A separation processing for industrial rare earth feed solution by phosphonium ionic liquid type saponification strategy, *J. Rare Earths* 35 (2017) 290–299.
- [57] Y. Dong, X. Sun, Y. Wang, C. Huang, Z. Zhao, The sustainable and efficient ionic liquid-type saponification strategy for rare earth separation processing, *ACS Sustain. Chem. Eng.* 4 (2015) 1573–1580.
- [58] S. Riaño, K. Binnemans, Extraction and separation of neodymium and dysprosium from used NdFeB magnets: an application of ionic liquids in solvent extraction towards the recycling of magnets, *GC 17* (2015) 2931–2942.
- [59] E. Padhan, K. Sarangi, Recovery of Nd and Pr from NdFeB magnet leachates with bi-functional ionic liquids based on Aliquat 336 and Cyanex 272, *Hydrometallurgy* 167 (2017) 134–140.
- [60] D. Hwang, T. Tsukahara, N. Miyamoto, Y. Ikeda, Spectrophotometric studies on solubility of lanthanoid(III) complexes with fluorinated β -diketonates in supercritical carbon dioxide, *J. Supercrit. Fluids* 110 (2016) 251–256.
- [61] M.D. Samsonov, T.I. Trofimov, Y.M. Kulyako, S.E. Vinokurov, D.A. Malikov, G. S. Batorshin, B.F. Myasoedov, Recovery of rare earth elements, uranium, and thorium from monazite concentrate by supercritical fluid extraction, *Radiochemistry* 57 (2015) 343–347.
- [62] C.C. Sara, O.F. José, Extraction techniques with deep eutectic solvents, *Trends Anal. Chem.* 105 (2018) 225–239.
- [63] J.E. Quinn, K.H. Soldenhoff, G.W. Stevens, Solvent extraction of rare earth elements using a bifunctional ionic liquid. Part 2: separation of rare earth elements, *Hydrometallurgy* 169 (2017) 621–628.
- [64] J.L. L. G. C. P.D.L.R. A. J.H. F. S. S. J. A. F. Recent advances in supported ionic liquid membrane technology, *J. Membr. Sci.* 376 (2011) 1–14.
- [65] J. Martínez, R. Rodríguez Varela, K. Forsberg, Å. Rasmuson, Factors influencing separation selectivity of rare earth elements in flat sheet supported liquid membranes, *Chem. Eng. Sci.* 191 (2018) 134–155.
- [66] S. B. A. G. F.I. A. Recent progress in ionic liquid membranes for gas separation, *J. Mol. Liq.* 266 (2018) 330–341.
- [67] P.V. Vernekar, Y.D. Jagdale, A.D. Sharma, A.W. Patwardhan, A.V. Patwardhan, S. A. Ansari, P.K. Mohapatra, Simultaneous extraction of neodymium and uranium using hollow fiber supported liquid membrane, *Sep. Sci. Technol.* 49 (2014) 1509–1520.
- [68] A.C. Ni, Am, Y. Wang, S. Chen, G. Chang, S. You, Simultaneous recovery of rare earth elements from waste permanent magnets (WPMs) leach liquor by solvent extraction and hollow fiber supported liquid membrane, *Chem. Eng. Process* 148 (2020), 107831.
- [69] A.H. Maad, A.M. Ahmed, A.A. Mohammed, Application of emulsion and Pickering emulsion liquid membrane technique for wastewater treatment: an overview, *Environ. Sci. Pollut. R.* 26 (2019) 36184–36204.
- [70] L. Chen, J. Chen, Asymmetric membrane containing ionic liquid [A336][P507] for the preconcentration and separation of heavy rare earth lutetium, *ACS Sustain. Chem. Eng.* 4 (2016) 2644–2650.
- [71] H. Siyan, C. Ji, C. Li, Z. Dan, L. Chuanying, A polymer inclusion membrane functionalized by di(2-ethylhexyl) phosphinic acid with hierarchically ordered porous structure for Lutetium(III) transport, *J. Membr. Sci.* 593 (2019), 117458.
- [72] W. Yitan, C. Li, Y. Yongsheng, C. Ji, D. Jiangdong, D. Xiaohui, Separation of adjacent heavy rare earth Lutetium (III) and Ytterbium (III) by task-specific ionic liquid Cyphos IL 104 embedded polymer inclusion membrane, *J. Membr. Sci.* 610 (2020), 118263.
- [73] M.Y. Byung, E.S. Jae, D.L. Hee, B.P. Ho, Graphene and graphene oxide membranes for gas separation applications, *Curr. Opin. Chem. Eng.* 16 (2017) 36–47.
- [74] H. Tan, X. Zhang, Z. Li, Q. Liang, J. Wu, Y. Yuan, S. Cao, J. Chen, J. Liu, H. Qiu, Nitrogen-doped nanoporous graphene induced by a multiple confinement strategy for membrane separation of rare earth, *iScience* 24 (2021), 101920.
- [75] F. Li, Z. Yang, H. Weng, G. Chen, M. Lin, C. Zhao, High efficient separation of U (VI) and Th(IV) from rare earth elements in strong acidic solution by selective sorption on phenanthroline diamide functionalized graphene oxide, *Chem. Eng. J.* 332 (2018) 340–350.
- [76] K. Cui, B. Gao, M. Tai, B. Su, A facile bionic strategy towards Gd (III)-imprinted membranes via interlaced stacking of one-dimensional/two-dimensional nanocomposite materials, *J. Taiwan Inst. Chem. E.* 95 (2019) 652–659.
- [77] X. Zheng, Y. Zhang, T. Bian, Y. Zhang, Z. Li, J. Pan, Oxidized carbon materials cooperative construct ionic imprinted cellulose nanocrystals films for efficient adsorption of Dy(III), *Chem. Eng. J.* 381 (2020), 122669.
- [78] Z. Li, Y. Liu, Y. Zhao, X. Zhang, L. Qian, L. Tian, J. Bai, W. Qi, H. Yao, B. Gao, J. Liu, W. Wu, H. Qiu, Selective separation of metal ions via monolayer nanoporous graphene with carboxyl groups, *Anal. Chem.* 88 (2016) 10002–10010.
- [79] Z. Li, X. Zhang, H. Tan, W. Qi, L. Wang, M.C. Ali, H. Zhang, J. Chen, P. Hu, C. Fan, H. Qiu, Combustion fabrication of nanoporous graphene for ionic separation membranes, *Adv. Funct. Mater.* 28 (2018), 1805026.
- [80] N.N. Hidayah, S.Z. Abidin, The evolution of mineral processing in extraction of rare earth elements using liquid-liquid extraction: a review, *Min. Eng.* 121 (2018) 146–157.
- [81] A. Abbasalizadeh, A. Malfliet, S. Seetharaman, J. Sietsma, Y. Yang, Electrochemical extraction of rare earth metals in molten fluorides: conversion of rare earth oxides into rare earth fluorides using fluoride additives, *J. Sust. Metall.* 3 (2017) 627–637.
- [82] L. Jiang, W. Zhang, C. Luo, D. Cheng, J. Zhu, Adsorption toward trivalent rare earth element from aqueous solution by zeolitic imidazolate frameworks, *Ind. Eng. Chem. Res.* 55 (2016) 6365–6372.
- [83] M. Majdan, A. Gladysz-Plaska, S. Pikus, D. Sternik, O. Maryuk, E. Zieba, P. Sadowski, Tetrad effect in the distribution constants of the lanthanides in their adsorption on the zeolite A, *J. Mol. Struct.* 702 (2004) 95–102.
- [84] M. Majdan, S. Pikus, A. Gladysz-Plaska, L. Fuks, E. Zieba, Adsorption of light lanthanides on the zeolite A surface, *Colloids Surf. A* 209 (2002) 27–35.
- [85] Y. Hu, J. Florek, D. Larivière, F. Fontaine, F. Kleitz, Recent advances in the separation of rare earth elements using mesoporous hybrid materials, *Chem. Rec.* 18 (2018) 1261–1276.
- [86] J.A. Mattocks, J.A. Cotruvo, Biological, biomolecular, and bio-inspired strategies for detection, extraction, and separations of lanthanides and actinides, *Chem. Soc. Rev.* 49 (2020) 8315–8334.
- [87] C. Huang, B. Huang, Y. Dong, J. Chen, Y. Wang, X. Sun, Efficient and sustainable regeneration of bifunctional ionic liquid for rare earth separation, *ACS Sustain. Chem. Eng.* 5 (2017) 3471–3477.
- [88] L. Ramrakhiani, A. Halder, A. Majumdar, A.K. Mandal, S. Majumdar, S. Ghosh, Industrial waste derived biosorbent for toxic metal remediation: mechanism studies and spent biosorbent management, *Chem. Eng. J.* 308 (2017) 1048–1064.
- [89] Y.C. Sze, S. Pau-Loke, F.L. Beng, C. Jo-Shu, C.L. Tau, New prospects for modified algae in heavy metal adsorption, *Trends Biotechnol.* 37 (2019) 1255–1268.
- [90] Takehiko Tsuruta, Selective accumulation of light or heavy rare earth elements using gram-positive bacteria, *Colloid Surf. B-Biointerfaces* 52 (2006) 117–122.
- [91] H. Moriwaki, H. Yamamoto, Interactions of microorganisms with rare earth ions and their utilization for separation and environmental technology, *Appl. Microbiol. Biot.* 97 (2013) 1–8.
- [92] A.M.S. Khalid, F.H. Mohammed, M. Hamed, W. Yuezhou, G. Feng, M.A. Ayman, F. Toyohisa, G. Eric, Nd(III) and Gd(III) sorption on mesoporous amine-functionalized polymer/SiO₂ composite, *Molecules* 26 (2021) 1049.
- [93] Y. Hu, L.C. Misal Castro, E. Drouin, J. Florek, H. Kählig, D. Larivière, F. Kleitz, F. Fontaine, Size-selective separation of rare earth elements using functionalized mesoporous silica materials, *ACS Appl. Mater. Interfaces* 11 (2019) 23681–23691.
- [94] G. Sandipta, S. Gautam, J. U. P. Sagar, Novel biodegradable polymeric flocculant based on polyacrylamide-grafted tamarind kernel polysaccharide, *Bioresour. Technol.* 101 (2010) 9638–9644.
- [95] S. Iftikhar, V. Srivastava, A. Casas, M. Sillanpää, Synthesis of novel GA-g-PAM/SiO₂ nanocomposite for the recovery of rare earth elements (REE) ions from aqueous solution, *J. Clean. Prod.* 170 (2018) 251–259.
- [96] E.F. Glen, L. Yuehe, F. Sandy, C.B. Jerome, W. Hong, K. Ken, K. Shelley, Actinide sequestration using self-assembled monolayers on mesoporous supports, *Environ. Sci. Technol.* 39 (2005) 1324–1331.
- [97] Q. Gao, J. Xie, Y. Shao, C. Chen, B. Han, K. Xia, C. Zhou, Ultrafast and high-capacity adsorption of Gd(III) onto inorganic phosphorous acid modified mesoporous SBA-15, *Chem. Eng. J.* 313 (2017) 197–206.
- [98] F. Zhang, K. Ma, Y. Li, Q. Ran, C. Yao, C. Yang, H. Yu, S. Hu, S. Peng, Selective separation of thorium from rare earths and uranium in acidic solutions by phosphorodiamidate-functionalized silica, *Chem. Eng. J.* 392 (2020), 123717.
- [99] S. Ravi, S. Zhang, Y. Lee, K. Kang, J. Kim, J. Ahn, W. Ahn, EDTA-functionalized KCC-1 and KIT-6 mesoporous silicas for Nd³⁺ ion recovery from aqueous solutions, *J. Ind. Eng. Chem.* 67 (2018) 210–218.
- [100] X. Xu, J. Zou, J. Teng, Q. Liu, X. Jiang, F. Jiao, J. Yu, X. Chen, Novel high-gluten flour physically cross-linked graphene oxide composites: Hydrothermal fabrication and adsorption properties for rare earth ions, *Ecotoxicol. Environ. Saf.* 166 (2018) 1–10.
- [101] C. Liao, X. Zhao, X. Jiang, J. Teng, J. Yu, Hydrothermal fabrication of novel three-dimensional graphene oxide-pentaerythritol composites with abundant oxygen-containing groups as efficient adsorbents, *Microchem. J.* 152 (2020), 104288.

- [102] J. Florek, F. Chalifour, F. Bilodeau, D. Larivière, F. Kleitz, Nanostructured hybrid materials for the selective recovery and enrichment of rare earth elements, *Adv. Funct. Mater.* 24 (2014) 2668–2676.
- [103] J. Florek, D. Larivière, H. Kählig, S.L. Fiorilli, B. Onida, F. Fontaine, F. Kleitz, Understanding selectivity of mesoporous silica-grafted diglycolamide-type ligands in the solid-phase extraction of rare earths, *ACS Appl. Mater. Inter.* 12 (2020) 57003–57016.
- [104] C.E.D. Cardoso, J.C. Almeida, C.B. Lopes, T. Trindade, C. Vale, E. Pereira, Recovery of rare earth elements by carbon-based, *Nanomater. Rev., Nanomater.* 9 (2019) 814.
- [105] D.L. Ramasamy, V. Puhakka, E. Repo, S. Ben Hammouda, M. Sillanpää, Two-stage selective recovery process of scandium from the group of rare earth elements in aqueous systems using activated carbon and silica composites: dual applications by tailoring the ligand grafting approach, *Chem. Eng. J.* 341 (2018) 351–360.
- [106] P. Sharma, A.K. Singh, V.K. Shahi, Selective adsorption of Pb(II) from aqueous medium by cross-linked chitosan-functionalized graphene oxide adsorbent, *ACS Sustain. Chem. Eng.* 7 (2019) 1427–1436.
- [107] Z. Pin, J. Meipeng, Z. Qi, X. Rongming, L. Ruiping, Z. Xiwang, L. Huijuan, A new paradigm of ultrathin 2D nanomaterial adsorbents in aqueous media: graphene and GO, MoS₂, MXenes, and 2D MOFs, *J. Mater. Chem. A* 7 (2019) 16598–16621.
- [108] M.A. Radwa, N.A. Hani, F.A. Ahmed, A.A. Ahmed, M.A. M, U. A, M. Mamoun, Z. Xiaodong, D. Joydeep, Rare earth ions adsorption onto graphene oxide nanosheets, *Solvent Extr. Ion.-Exchange* 35 (2017) 91–103.
- [109] N.K. Gupta, B.C. Choudhary, A. Gupta, S.N. Achary, A. Sengupta, Graphene-based adsorbents for the separation of f-metals from waste solutions: a review, *J. Mol. Liq.* 289 (2019), 111121.
- [110] X. Zhao, X. Jiang, D. Peng, J. Teng, J. Yu, Behavior and mechanism of graphene oxide-tris(4-aminophenyl)amine composites in adsorption of rare earth elements, *J. Rare Earth.* 39 (2020) 90–97.
- [111] P.S. Richard, K.S. Scott, D.H. John, D.R. Robin, Dissolution of cellulose [correction of cellose] with ionic liquids, *J. Am. Chem. Soc.* 124 (2002) 4974–4975.
- [112] X. Xu, X. Jiang, F. Jiao, X. Chen, J. Yu, Tunable assembly of porous three-dimensional graphene oxide-corn zein composites with strong mechanical properties for adsorption of rare earth elements, *J. Taiwan Inst. Chem. E.* 85 (2018) 106–114.
- [113] X. Xu, J. Zou, X. Zhao, X. Jiang, F. Jiao, J. Yu, Q. Liu, J. Teng, Facile assembly of three-dimensional cylindrical egg white embedded graphene oxide composite with good reusability for aqueous adsorption of rare earth elements, *Colloid Surf. A-Physicochem. Eng. Asp.* 570 (2019) 127–140.
- [114] F.A. Ahmed, N.A. Hani, M.A. Radwa, Z. Xiaodong, F. Kerstin, Hierarchical porous zeolitic imidazolate frameworks nanoparticles for efficient adsorption of rare-earth elements, *Microporous Mesoporous Mater.* 278 (2018) 175–784.
- [115] K. Tu, B. Puértolas, M. Adobes Vidal, Y. Wang, J. Sun, J. Traber, I. Burgert, J. Pérez Ramírez, T. Keplinger, Green synthesis of hierarchical metal-organic framework/wood functional composites with superior mechanical properties, *Adv. Sci.* 7 (2020), 1902897.
- [116] Z. Lin, S. Daopeng, X. Chao, D. Xing, A.S. Mark, L. Jie, L. Peng, W. Yaxing, W. Yanlong, C. Lanhua, X. Chengliang, C. Jing, Z. Ruhong, Z. Chao, K.F. Omar, C. Zhifang, E.A. Thomas, W. Shuao, Identifying the recognition site for selective trapping of 99TcO₄⁻ in a hydrolytically stable and radiation resistant cationic metal-organic framework, *J. Am. Chem. Soc.* 139 (2017) 14873–14876.
- [117] Z. Li, J. Yang, K. Sui, N. Yin, Facile synthesis of metal-organic framework MOF-808 for arsenic removal, *Mater. Lett.* 160 (2015) 412–414.
- [118] H.E. Emam, R.M. Abdelhameed, H.B. Ahmed, Adsorptive performance of MOFs and MOF containing composites for clean energy and safe environment, *J. Environ. Chem. Eng.* 8 (2020), 104386.
- [119] W. Xu, C. Yu, L. Liu, Y. Zhai, R. Xu, H. Hou, An O- modified coordination polymer for rapid and selective adsorption of rare earth elements from aqueous solution, *Colloid Surf. A-Physicochem. Eng. Asp.* 607 (2020), 125464.
- [120] V. Kavun, M.A. van der Veen, E. Repo, Selective recovery and separation of rare earth elements by organophosphorus modified MIL-101(Cr), *Microporous Mesoporous Mater.* 312 (2021), 110747.
- [121] D.D. Jeroen, D.C. Jeriffa, V. Pieter, V.D.V. Pascal, Functionalized metal-organic-framework CMPO@MIL-101(Cr) as a stable and selective rare earth adsorbent, *J. Mater. Sci.* 51 (2016) 5019–5026.
- [122] J. De Decker, J. Rochette, J. De Clercq, J. Florek, P. Van Der Voort, Carbamoylmethylphosphine oxide-functionalized MIL-101(Cr) as highly selective uranium adsorbent, *Anal. Chem.* 89 (2017) 5678–5682.
- [123] J. Troyano, C. Çamur, L. Garzón-Tovar, A. Carné-Sánchez, I. Imaz, D. Maspoch, Spray-drying synthesis of MOFs, COFs, and related composites, *Acc. Chem. Res.* 53 (2020) 1206–1217.
- [124] Y.J. Tong, L.D. Yu, J. Zheng, G. Liu, Y. Ye, S. Huang, G. Chen, H. Yang, C. Wen, S. Wei, J. Xu, F. Zhu, J. Pawliszyn, G. Ouyang, Graphene oxide-supported lanthanide metal-organic frameworks with boosted stabilities and detection sensitivities, *Anal. Chem.* 92 (2020) 15550–15557.
- [125] W. Wei, Z. Liu, R. Wei, G. Han, C. Liang, Synthesis of MOFs/GO composite for corrosion resistance application on carbon steel, *RSC Adv.* 10 (2020) 29923–29934.
- [126] D. Yin, G. Huang, Q. Sun, Q. Li, X. Wang, D. Yuan, C. Wang, L. Wang, RGO/Co 3 O 4 composites prepared using GO-MOFs as precursor for advanced lithium-ion batteries and supercapacitors electrodes, *Electrochim. Acta* 215 (2016) 410–419.
- [127] D. Tian-Wei, Y. Bing, Hybrids based on lanthanide ions activated yttrium metal-organic frameworks: functional assembly, polymer film preparation and luminescence tuning, *J. Mater. Chem. C* 2 (2014) 5098–5104.
- [128] S. Feng, X. Du, M. Bat-Amgalan, H. Zhang, N. Miyamoto, N. Kano, Adsorption of REEs from aqueous solution by EDTA-chitosan modified with zeolite imidazole framework (ZIF-8), *Int. J. Mol. Sci.* 22 (2021), <https://doi.org/10.3390/ijms22073447>.
- [129] J. Wu, Z. Li, H. Tan, S. Du, T. Liu, Y. Yuan, X. Liu, H. Qiu, Highly selective separation of rare earth elements by Zn-BTC Metal-organic Framework/nanoporous Graphenevia In Situ Green Synthesis, *Anal. Chem.* 93 (2020) 1732–1739.
- [130] X.H. Xiong, Y. Tao, Z.W. Yu, L.X. Yang, L.J. Sun, Y.L. Fan, F. Luo, Selective extraction of thorium from uranium and rare earth elements using sulfonated covalent organic framework and its membrane derivate, *Chem. Eng. J.* 384 (2020), 123240.
- [131] J. Liu, L. Zeng, S. Liao, X. Liao, J. Liu, J. Mao, Y. Chen, T. Qiu, S. Ren, Highly efficient enrichment and adsorption of rare earth ions (yttrium(III)) by recyclable magnetic nitrogen functionalized mesoporous expanded perlite, *Chin. Chem. Lett.* 31 (2020) 2849–2853.
- [132] D. Dupont, W. Brullot, M. Bloemen, T. Verbiest, K. Binnemans, Selective uptake of rare earths from aqueous solutions by EDTA-functionalized magnetic and nonmagnetic nanoparticles, *ACS Appl. Mater. Inter.* 6 (2014) 4980–4988.
- [133] S.D.N. Almeida, H.E. Toma, Lanthanide ion processing from monazite based on magnetic nanohydrometallurgy, *Hydrometallurgy* 189 (2019), 105138.
- [134] X. Min, W. Yang, Y.F. Hui, C.Y. Gao, S. Dang, Z.M. Sun, Fe₃O₄@ZIF-8: a magnetic nanocomposite for highly efficient UO₂(2+) adsorption and selective UO₂(2+)/Ln(3+) separation, *Chem. Commun.* 53 (2017) 4199–4202.
- [135] X. Yang, D.K. Debeli, G. Shan, P. Pan, Selective adsorption and high recovery of La³⁺ using graphene oxide/poly (N-isopropyl acrylamide-maleic acid) cryogel, *Chem. Eng. J.* 379 (2020), 122335.
- [136] M. Wang, X. Li, W. Hua, L. Shen, X. Yu, X. Wang, Electrospun poly(acrylic acid)/silica hydrogel nanofibers scaffold for highly efficient adsorption of lanthanide ions and its photoluminescence performance, *ACS Appl. Mater. Inter.* 8 (2016) 23995–24007.
- [137] S. Zhu, M. Xia, Y. Chu, M.A. Khan, W. Lei, F. Wang, T. Muhmood, A. Wang, Adsorption and desorption of Pb(II) on L-lysine modified montmorillonite and the simulation of interlayer structure, *Appl. Clay Sci.* 169 (2019) 40–47.
- [138] S. Zhu, M.A. Khan, F. Wang, Z. Bano, M. Xia, Exploration of adsorption mechanism of 2-phosphonobutane-1,2,4-tricarboxylic acid onto kaolinite and montmorillonite via batch experiment and theoretical studies, *J. Hazard. Mater.* 403 (2021), 123810.
- [139] S. Zhu, M.A. Khan, F. Wang, Z. Bano, M. Xia, Rapid removal of toxic metals Cu²⁺ and Pb²⁺ by amino trimethylene phosphonic acid intercalated layered double hydroxide: a combined experimental and DFT study, *Chem. Eng. J.* 392 (2020), 123711.
- [140] H. Fang, B.E. Cole, Y. Qiao, J.A. Bogart, T. Cheisson, B.C. Manor, P.J. Carroll, E. J. Schelter, Electro-kinetic separation of rare earth elements using a redox-active ligand, *Angew. Chem. Int. Ed.* 56 (2017) 13450–13454.
- [141] V. Haley, E. Johannes, S. Xiao, Redox-copolymers for the recovery of rare earth elements by electrochemically regenerated ion-exchange, *J. Mater. Chem. A* 9 (2021) 20068–20077.
- [142] C. Li, D.L. Ramasamy, M. Sillanpää, E. Repo, Separation and concentration of rare earth elements from wastewater using electrodialysis technology, *Sep. Purif. Technol.* 254 (2021), 117442.
- [143] R. Gueroult, J. Rax, N.J. Fisch, Opportunities for plasma separation techniques in rare earth elements recycling, *J. Clean. Prod.* 182 (2018) 1060–1069.